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# Crown-Substituted Phthalocyanines—Components of Molecular Ionelectronic Materials and Devices<sup>1</sup>

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**Abstract**—This review summarizes the synthetic approaches towards oxa-crown-substituted phthalocyanines (CRPc's) including various metal complexes of *s*-, *p*-, *d*-, and *f*-elements. The template cyclotetramerization of phthalonitriles and direct metalation pathways for synthesis of target compounds are discussed and compared. It is demonstrated that obtained results have contributed to coordination, supramolecular, and materials chemistry of phthalocyanines in general. An analysis of the results published to date reveals the challenges of these compounds for their use in high-tech applications.

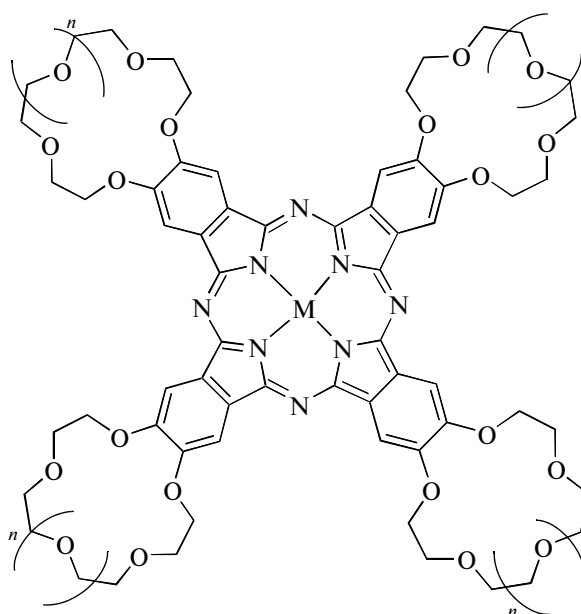
**Keywords:** phthalocyanine, crown-ether, supramolecular chemistry, cation-induced assembling, ionoelectronics

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Tetrapyrrolic macrocycles can form supramolecular assemblies in solution themselves via dynamic non-covalent interactions, resulting in the strong tendency of these compounds toward self-aggregation, particularly due to the stacking interaction between their huge aromatic systems [1]. These interactions also play a significant role in the ordering of molecules in solid state and predetermining therefore special properties of materials for example an electronic conductivity. From this viewpoint, the control over the aggregation of tetrapyrroles by means of supramolecular chemistry is a remarkable development [2–8], since it affords insight into the effects which arise from supramolecular assembling and therefore gives novel approaches towards material with outstanding physical-chemical characteristics. To achieve this control, additional binding sites should be introduced at the periphery of the tetrapyrrolic macrocycle. Variation of distinct binding centers of the resulting composite molecule–coreceptor–affords fine tuning of specific complexation of the entire receptor. With this aim, numerous examples of porphyrins with additional binding sites have been synthesized and studied, revealing their promising potential in novel high-tech applications [9].

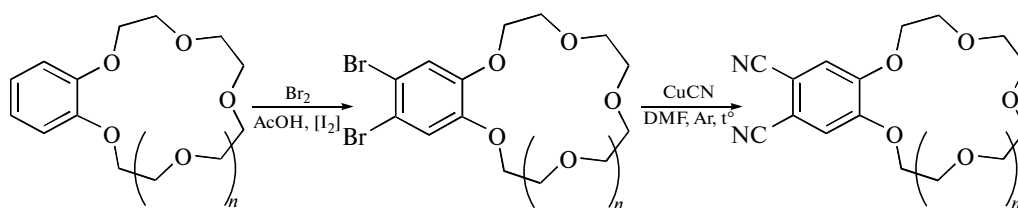
At the same time, heterotopic phthalocyanines have been less investigated. Nevertheless, this class of compounds includes such bright representatives as

crown-substituted phthalocyanines (CRPc) [10]. In this review we focus our attention on synthesis of oxa-crown-substituted compounds, appended directly to  $\beta, \beta'$ -positions of the phthalocyanine core—both



**Fig. 1.** Tetra-crown-phthalocyanines, ( $n = 1$ )  $M[(15C5)_4Pc]$ , ( $n = 2$ )  $M[(18C6)_4Pc]$ , ( $n = 3$ )  $M[(21C7)_4Pc]$ ,  $M = 2H$  and various metals.

<sup>1</sup> The article is published in the original.



**Scheme 1.** Synthesis of dibromobenzo- and dicyanobenzo-crown-ethers [23–25, 29].

ligands and metal complexes (Fig. 1). Introduction of crown-ethers as integral part of phthalocyanine molecule results in new abilities to supramolecular assembling due to binding of alkali- and alkaline metal cations together with ammonia ions. Other examples of CRPc's contain aza- and thia-ether fragments are also described as well as more complicated macrocyclic chelating systems [11–19], which can bind transition metal ions. Since the properties of crown-phthalocyanines can be modulated by interaction with cations, these materials and devices relate to the challenging field of ionoelectronics.

Remarkably, the synthesis of oxa-crown-substituted phthalocyanines was first reported independently and almost simultaneously in the example of copper tetra-15-crown-5-phthalocyaninate,  $\text{Cu}[(15\text{C}5)_4\text{Pc}]$  by three research groups headed by Bekâroğlu [20], Kobayashi [21] and Nolte [22] in 1986, with subsequent full papers [23–25].

Notably, all three groups had different suppositions that led them to discover the potential fields of application of CRPc's.

Bekâroğlu and coworkers started from the assumption that making crown-ether units an integral part of the Pc molecule would improve its solubility [20]. This assumption was based on the observation by Ziolo and coworkers, who observed the increasing solubility of  $\text{Na}_2\text{Pc}$  and  $\text{K}_2\text{Pc}$  after interaction with macrocyclic polyethers as complexing ligands for alkali ions [26, 27]. Kobayashi and Nishiyama induced aggregation of  $\text{Cu}[(15\text{C}5)_4\text{Pc}]$  using  $\text{K}^+$  cations, obtaining thereby an attractive colorimetric reagent for concentrations of this cation as low as  $10^{-8}$  M [21]. They demonstrated that the solubility of  $\text{Cu}[(15\text{C}5)_4\text{Pc}]$  in common organic solvents decreased in the order:  $\text{CHCl}_3$  (8.3 mM) >  $\text{CH}_2\text{Cl}_2$  (6.1 mM)  $\gg$  DMF, DMSO, toluene, benzene, THF, and EtOAc (<0.01 mM). Nolte and coworkers proposed  $\text{Cu}[(15\text{C}5)_4\text{Pc}]$  as a heterotopic ligand for multifunctional catalysis and carrier systems [22].

All authors investigated the affinity of  $\text{Cu}[(15\text{C}5)_4\text{Pc}]$  toward alkali cations and concluded that this new coreceptor may be utilized as a powerful and selective colorimetric sensor and extracting reagent towards these ions.

To investigate the correlations between the structures of coreceptors, the processes of their cation-

induced assembly and the physical-chemical characteristics of the corresponding monomers and aggregates, numerous other CRPc's have been prepared and extensively investigated over the last two decades.

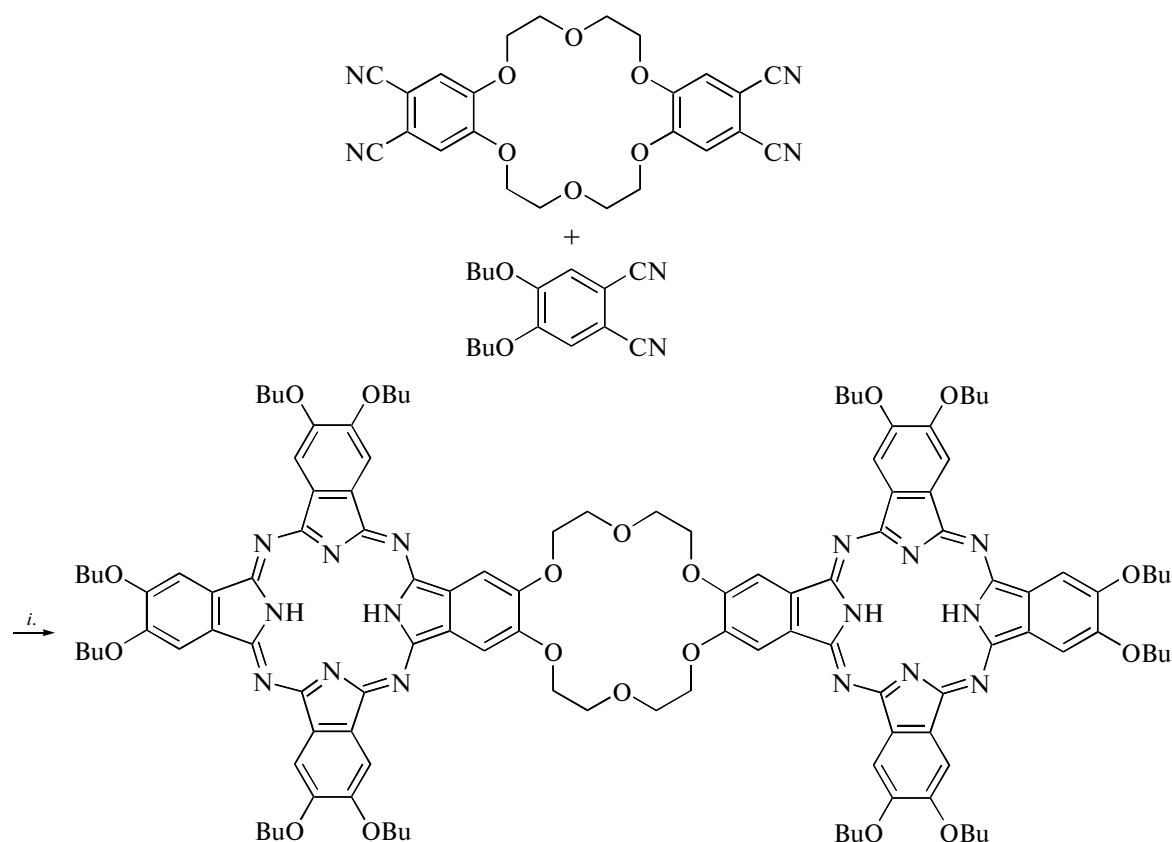
## 1. SYNTHESIS AND PROPERTIES OF CROWN-SUBSTITUTED PHTHALOCYANINES

### 1.1. Synthesis of Precursors

The first crown-phthalocyanine,  $\text{Cu}[(15\text{C}5)_4\text{Pc}]$ , was prepared in two steps, starting with the commercially available benzo-15-crown-5. Its bromination gave 4',5'-dibromobenzo-15-crown-5 (DBB15C5), which was treated with a large excess of CuCN in boiling quinoline or pyridine [20, 25], or in DMF/pyridine mixture [22], yielding greenish-blue  $\text{Cu}[(15\text{C}5)_4\text{Pc}]$  in ~50% yield as a product of template cyclotetramerization of the intermediate 4',5'-dicyanobenzo-15-crown-5 (DCB15C5). In order to obtain other metal complexes, as well as metal-free ligand, the control over cyanation conditions should be very accurate to avoid the conversion of forming DCB15C5 into corresponding copper complex. Therefore, the accessibility of dicyanobenzo-crown precursors is the key factor defining an efficient approach toward corresponding CRPc ligands.

Rosenmund-Braun cyanation of DBB15C5 with three equiv of CuCN in refluxing DMF gave DCB15C5 (Scheme 1) with the yield of the compound varying from 31–68% as reported by different authors [23–25]. It was shown that this reaction is particularly sensitive to inert atmosphere, purity of the DMF and reaction time [28]. Using a similar methodology, dicyano-derivatives of benzo-12-crown-4, 18-crown-6 and 21-crown-7 were prepared [24].

Together with benzo-crown-ethers, cyano-derivatives of dibenzo-crown-ethers were synthesized. To obtain symmetrical tetracyano-dibenzo-18-crown-6 (TCDB18C6) [30, 31] and tetracyano-dibenzo-24-crown-8 (TCDB24C8) [32], bromination of unsubstituted dibenzo-crowns was performed with subsequent cyanation. In the presence of excess CuCN, the formation of polymeric complexes was observed via template condensation of the intermediate tetracyano-dibenzo-crown-ether [30]. Formation of polymeric copper phthalocyaninate could be avoided



**Scheme 2.** Synthesis of binuclear phthalocyanine sharing one crown-ether ring. Reaction conditions: *i.* DMAE,  $\text{NH}_3$ , reflux, 8 hours, yield 6.6% [35].

when Pd-catalyzed cyanation of tetrabromo-dibenzo-24-crown-8 was performed [33].

To prepare metal CRPc's two general methodologies were applied—*template* cyclotetramerization of phthalonitriles in the presence of metal salts and *direct* metalation of the corresponding free base phthalocyanines, synthesized beforehand. All these methods will be discussed in following sections.

### 1.2. Synthesis of Metal-Free Crown-Phthalocyanines

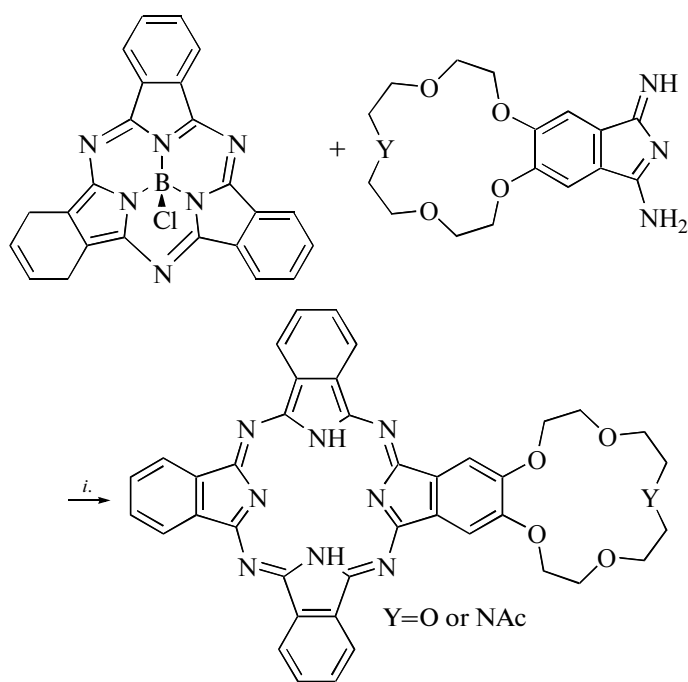
Common methods of phthalonitrile cyclotetramerization can be applied for synthesis of metal free CRPc's. Free base tetra-15-crown-5-phthalocyanine  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  was firstly obtained in 26% yield by refluxing DCB15C5 in *N,N*-dimethylaminoethanol (DMAE) for 17 hours [24]. Passing ammonia gas through the solution afforded an improved yield of 38% [23]. The same ligand was synthesized by fusion of the mixture of DCB15C5 and hydroquinone in a sealed tube at  $180^\circ\text{C}$  for 16 hours [25]. The yield of  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  was 32.2%. Tetra-18-crown-6-phthalocyanine  $\text{H}_2[(18\text{C}6)_4\text{Pc}]$  could be prepared in 12% in DMAE and in 7% yield by using the “fusion” method [34].

Cross-condensation of TCDB18C6 with 4,5-dibutoxyphthalonitrile in DMAE under a stream of ammonia gas yielded a binuclear compound, where two Pc units are bridged by 18-crown-6 unit (Scheme 2) [35]. Long-range electronic interaction between chromophoric units was evidenced by UV-Vis and MCD studies.

To make low-symmetry ligands containing only one macrocyclic substituent, the ring expansion of SubPc by crown-substituted diiminoisoindoline was used (Scheme 3) [36]. However, low solubility of the resulting compounds limited their further investigation.

### 1.3. Template Synthesis of *d*-Metal Crown-Phthalocyanines

The complexes  $\text{M}[(15\text{C}5)_4\text{Pc}]$ ,  $\text{M} = \text{Fe}, \text{Co}, \text{Ni}$  were prepared in 26–51% yield by refluxing DCB15C5 in ethyleneglycol in the presence of an appropriate metal salt [23, 37]. Another protocol to prepare the Co(II) and Ni(II) complexes involves refluxing of metal chlorides mixed with DCB15C5 in ethyleneglycol and quinoline, respectively. The complexes were obtained in 46 and 50% yields. The Fe(II)



**Scheme 3.** Sub-Pc ring expansion reaction, applied to prepare CRPc's, bearing single macrocyclic substituent. Reaction conditions: *i.*—DMSO/1-CN 2 : 1 vol, 12 hours, yields for Y = O, 44.5%, for Y = NAc, 65% [36].

complex was prepared in 16% yield via the reaction of DCB15C5 and  $\text{Fe}(\text{CO})_5$  in ethyleneglycol [25].

Template condensation of TCDB18C6 in the presence of  $\text{MCl}_2$  ( $\text{M} = \text{Cu}, \text{Ni}$ ) in quinoline yielded network-like polymeric compounds in 68 and 76% yields, respectively (Fig. 2). In the case of cobalt, addition of ammonium molybdate was required to obtain the polymer (51%); otherwise only the bisimide derivative of the starting nitrile was obtained. The Fe(II) complex (43%) was prepared by reaction of TCDB18C6 with  $\text{Fe}(\text{CO})_5$  in ethyleneglycol. Terminal cyano-groups could be converted into imido-groups by hydrolysis in sulfuric acid. The obtained polymers were found to be good extracting agents for alkali metal ions as well as for silver (although the later is complexed almost irreversibly) [30, 31].

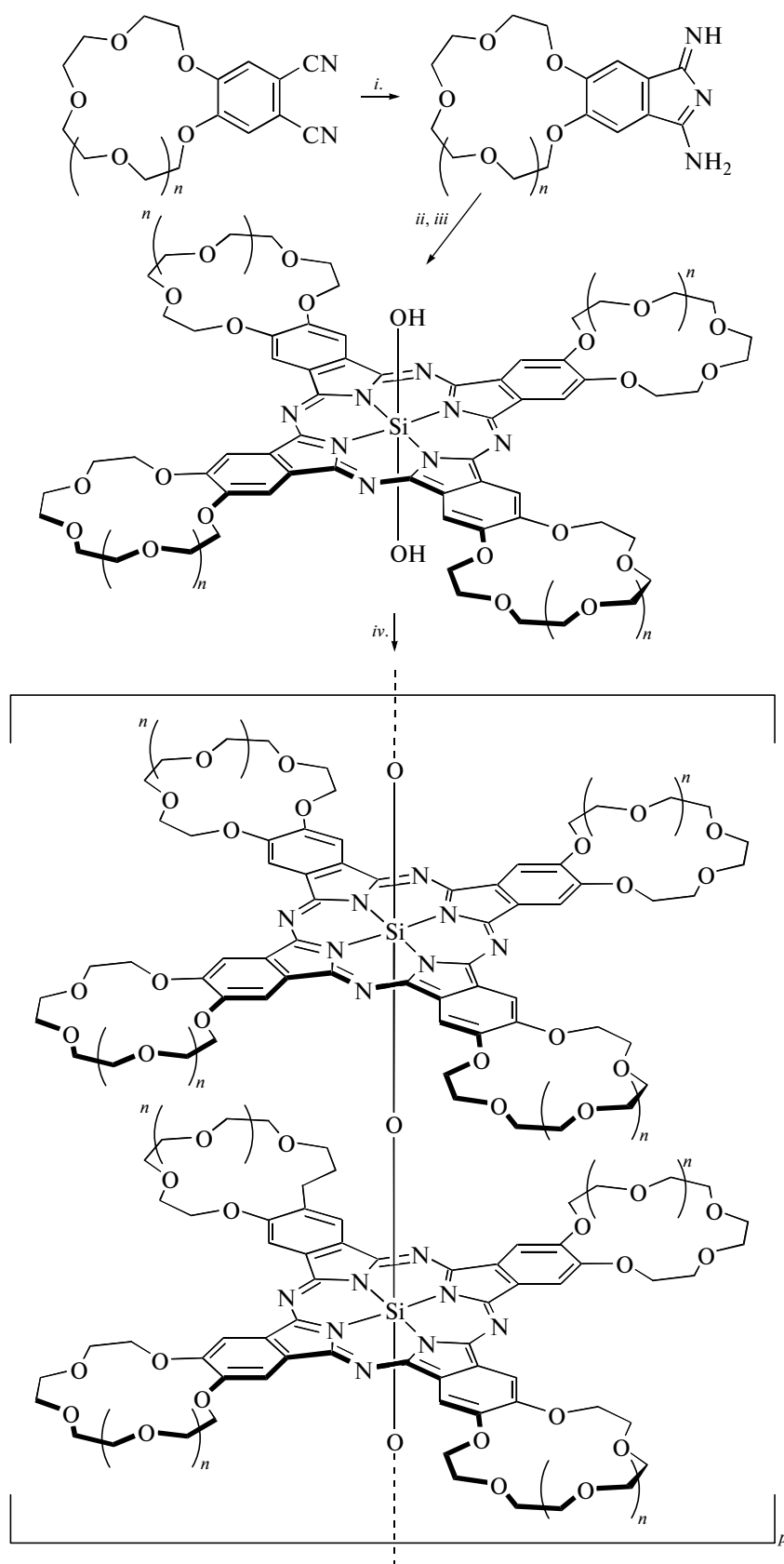
A template reaction was also used to prepare Pb(II) complex. Interestingly,  $\text{Pb}[(15\text{C}5)_4\text{Pc}]$  could be prepared only by solvent-free interaction of PbO with DBC15C5 (yield, 19%) [38]. The reaction performed in ethyleneglycol produced only  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$ , probably because of oxidation of Pb(II) to Pb(IV), which supplies the necessary two electrons to complete the 18- $\pi$  electron system of the phthalocyanine moiety. Similarly, treating TCDB18C6 with PbO in ethyleneglycol yielded 15% of metal free polymeric CRPc (Fig. 2,  $\text{M} = 2\text{H}$ ) [31].

Polysiloxane crown-phthalocyanines were proposed as compounds, forming ionic channels, due to formation of stacked quasi-1D arrangement of CRPc's which is provided by a linear polysiloxane

backbone. Such compounds, when incorporated in vesicle bilayers, could be used as synthetic models to mimic both electron and ion channels [39, 40].

To make these polymers, the following sequence of steps was taken (Scheme 4): crown-substituted phthalonitriles (from 15C5 to 21C7) were converted into the corresponding 1,3-diiminoisoindolines by passing ammonia gas through a methanol solution in the presence of MeONa. The obtained compounds reacted with  $\text{SiCl}_4$  in quinoline at  $130^\circ\text{C}$  to produce the corresponding dichloro-silicon CRPc's, which were then hydrolyzed to dihydroxysilicon complexes. Their polycondensation was performed in quinoline at  $200^\circ\text{C}$  in the presence of  $\text{CaCl}_2$ . Heating for two to five hours yielded a mixture of soluble dimer and trimer (dimer could be isolated by chromatography on Sephadex LH60), while prolonged heating (70 hours) yielded insoluble polymers. Similar protocol was simultaneously used in [38] to polymerize  $[(15\text{C}5)_4\text{Pc}]\text{Si}(\text{OH})_2$  in refluxing quinoline (b.p. of quinoline is  $238^\circ\text{C}$ ). Such increase of temperature afforded polymers with the average degree of polymerization equal to 25 units per chain. It was estimated by quantitatively determining the relative intensity of the band at  $870\text{ cm}^{-1}$  (O—Si—O antisymmetric stretching) in the IR spectra of monomer and polymer relative to the well-characterized invariant Ar—O—C modes of the crown ether moieties at  $1210\text{ cm}^{-1}$ .

Boron-templated condensation was used to prepare crown-substituted subphthalocyanines (Fig. 3) by reaction of DCB15C5, either with  $\text{BCl}_3$  in 1-CN



**Scheme 4.** Synthesis of polymeric crown-phthalocyaninato-polysiloxanes. Reaction conditions: *i.*  $\text{NH}_3$ ,  $\text{MeONa}$ ,  $\text{MeOH}$ ,  $95^\circ\text{C}$ , 4 hours (yields  $n = 1.83\%$ ,  $n = 2.74\%$ ,  $n = 3.56\%$ ); *ii.*  $\text{SiCl}_4$ , quinoline,  $100^\circ\text{C}$ , 12 hours; *iii.* Water, room temperature, 4 hours (yields  $n = 1.30\%$ ,  $n = 2.25\%$ ,  $n = 3.32\%$ ); *iv.*  $\text{CaCl}_2$ , quinoline,  $200^\circ\text{C}$ , 70 hours (yields  $n = 1.80\%$ ,  $n = 2.69\%$ ,  $n = 3.71\%$ ) [40].

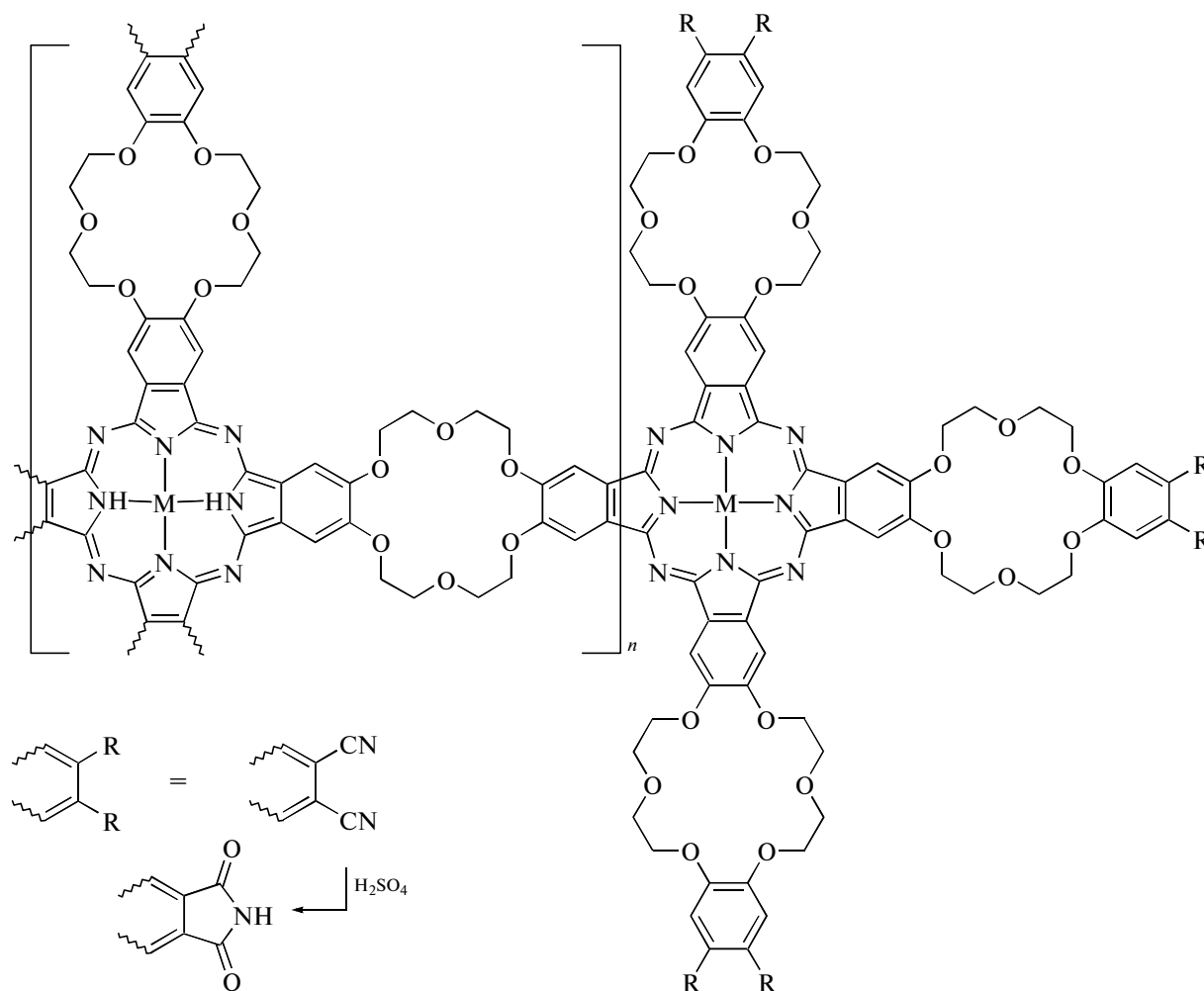


Fig. 2. Polymeric phthalocyanines, prepared from TCDB18C6 ( $M = 2H, Fe, Co, Ni, Cu$ ) [30, 31].

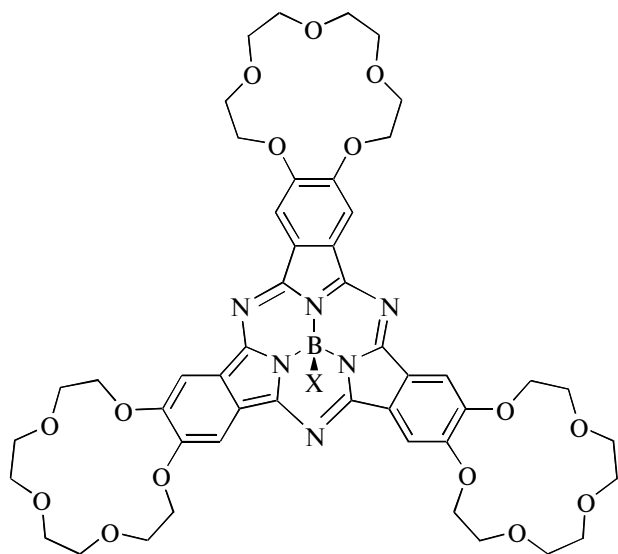


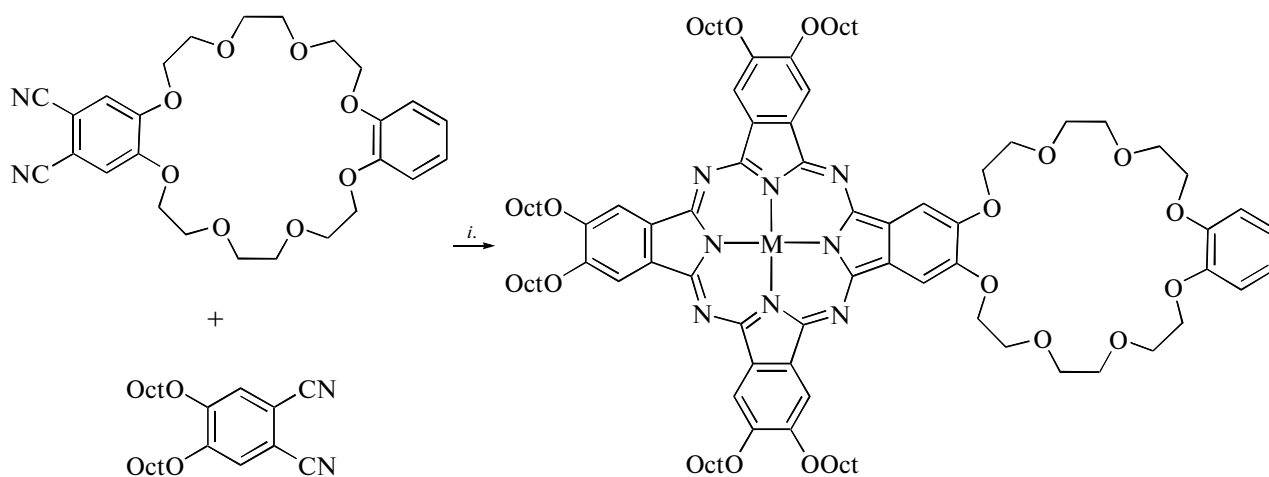
Fig. 3. Crown-substituted boron subphthalocyanines [(15C5)<sub>3</sub>SubPc]BX,  $X = Cl$  [41] or  $Ph$  [42].

(yield, 3.5%) [41] or with  $BPh_3$  in naphthalene in the presence of DBU (yield, 0.4%) [42]. This low yield was ascribed to the difficulty of the elimination of two phenyl groups from  $BPh_3$ .

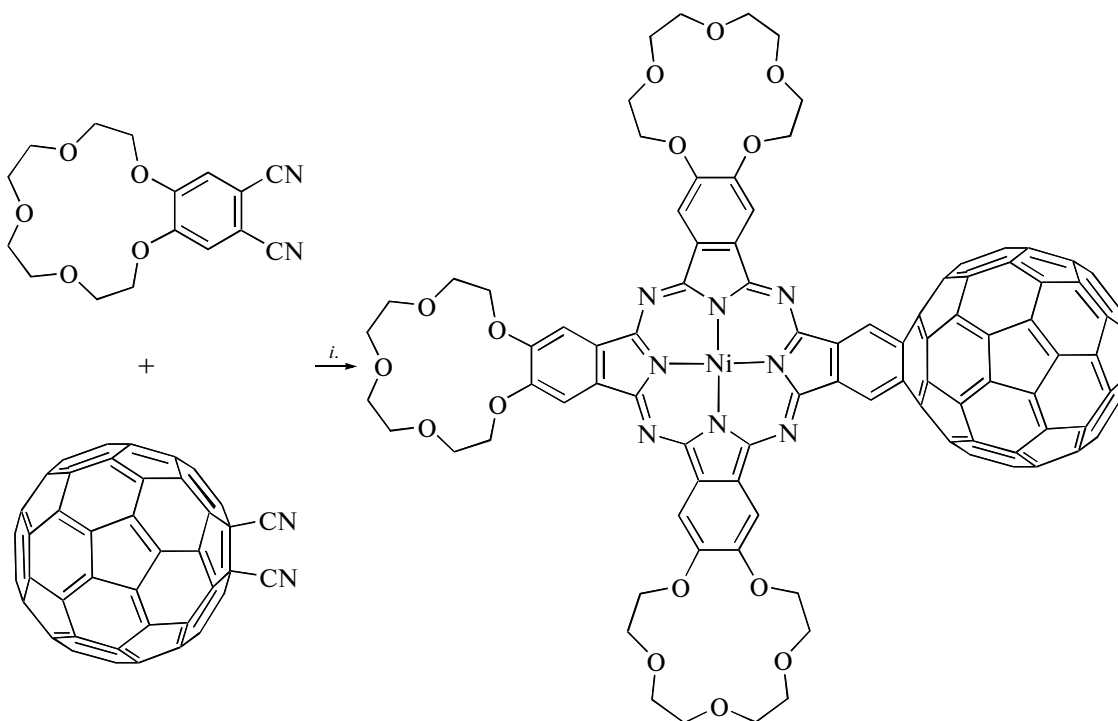
Template cross-condensation is a commonly used approach to prepare low-symmetry compounds [43]. For example, the template reaction was used to prepare low-symmetry  $Zn(II)$ ,  $Ni(II)$  and  $Cu(II)$  CRPc's, starting from DCDB24C8 and 4,5-dioctyloxyphthalonitrile (Scheme 5) [17]. This molecule was used as a building-block for preparation of photoactive  $Pc-C_{60}$  assembly with pseudorotaxane topology.

Under similar conditions, cross-condensation of TCDB24C8 with 4,5-dioctyloxyphthalonitrile in the presence of  $Zn(OAc)_2$  yielded 10% of a binuclear complex  $Zn_2[(24C8)(OctO)_{12}Pc_2]$  with one bridging crown-ether ring (the ligand is similar to that given in Scheme 2) [32].

Reaction of TCDB24C8 with 4,5-dibutoxyphthalonitrile in the presence of  $Mg(OAc)_2$  in refluxing mixture of isoamyl alcohol and chlorobenzene afforded



**Scheme 5.** Synthesis of mono-24-crown-8-phthalocyaninates  $M[(B24C8)(OctO)_6Pc]$ : Reaction conditions: *i.*  $M(OAc)_2$ , DMAE, reflux, 24 hours, yields  $M = Ni$  16%,  $Zn$  30%,  $Cu$  20% [17].

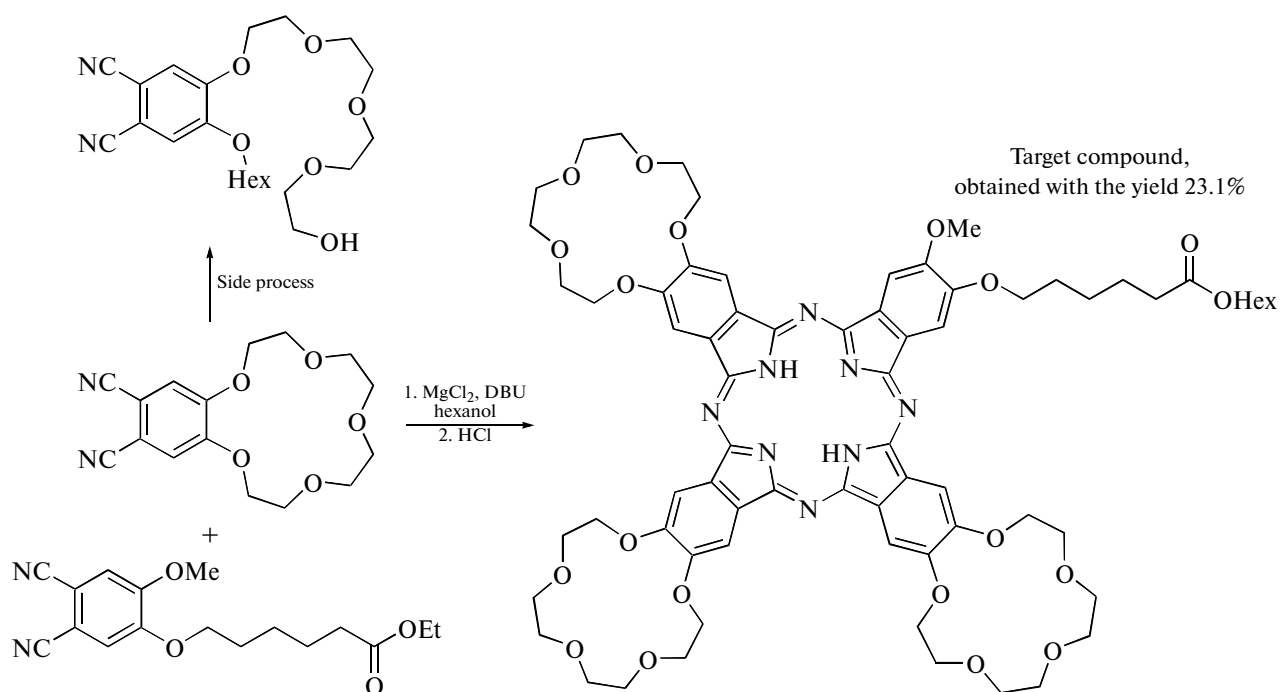


**Scheme 6.** Synthesis of 15-crown-5-substituted tribenzotetraazachlorin- $C_{60}$  conjugate. Reaction conditions: *i.*  $NiCl_2$ ,  $(NH_4)_2MoO_4$ , quinoline, reflux, 1 hour, yield 4.3% [45].

binuclear phthalocyaninate, sharing one common 24-crown-8-ring in 39% yield. Its demetalation in  $CHCl_3/CF_3COOH$  mixture afforded corresponding metal-free phthalocyanine, which could be subsequently metalated with nickel acetate [44]. Such binuclear phthalocyanines are supposed to act as pincer receptors for bifunctional substrates due to conformational flexibility of 24-crown-8 fragment.

The unique example of the complex containing three crown-ether units and one fullerene ball, directly appended to tetrapyrrole scaffold, was prepared by cross-condensation of DCB15C5 and dicyanofullerene, templated by  $Ni(II)$  ions (Scheme 6) [45]. The spectral characteristics (UV-Vis and MCD) of the product conjugate indicate that a large degree of MO mixing occurs between the tetrapyrrolic core and  $C_{60}$  moieties in the ground state.





**Scheme 7.** Synthesis of low-symmetry phthalocyanine bearing three crown-ether groups and ester chain. Possible side process—crown-ether nucleophilic cleavage, is shown [47].

Low-symmetry CRPc's bearing carboxylic terminated side chains were prepared to graft these compounds onto surfaces [46, 47]. To make these compounds, Mg(II)-templated DBU-promoted cross-condensations of DCB15C5 and ethyl-hexanoate substituted phthalonitrile [46] in 1-hexanol were performed [48, 47]. Demetalation of the resulting magnesium complexes yielded the mixture of corresponding ligands. The target compound with three crown-ether rings and one ester chain was isolated by preparative TLC in 23.1% yield (Scheme 7). The overall yield of this ligand together with other possible phthalocyanines was estimated to be around 64%. Two side reactions were observed, namely transesterification of hexanoate esters with replacement of ethyl by hexyl and, unexpectedly, partial crown-ether cleavage reaction with the formation of an open-chain derivative. The cleavage was facilitated by magnesium ions and more significantly, by calcium ions once they were added to the reaction mixture.

Similar approach was used for synthesis of well-soluble low-symmetry complex, bearing only one crown-ether macrocycle. For this purpose, Mg-templated condensation of DCB15C5 and dibutoxyphthalonitrile was performed in refluxing isoamyl alcohol [49]. Chromatographic separation of products afforded target complex  $\text{Mg}[(15\text{C}5)(\text{BuO})_6\text{Pc}]$  in 41% yield. Its demetalation in mixture of DMF and aq. HCl afforded corresponding ligand in 82% yield. Aggregation of  $\text{H}_2[(15\text{C}5)(\text{BuO})_6\text{Pc}]$  in solution was studied by UV-Vis spectroscopy.

#### 1.4. Direct Synthesis of *d*-Metal Crown-Phthalocyaninates

Symmetrical Zn(II) CRPc's were synthesized by direct metalation of the metal-free ligand. Refluxing of  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  with  $\text{Zn}(\text{OAc})_2$  in  $\text{C}_2\text{H}_4\text{Cl}_2/\text{EtOH}$  for several days in the dark yielded 78% of the complex [23]. The same complex was prepared when  $\text{Zn}(\text{acac})_2$  reacted with  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  in refluxing 1,2,4-trichlorobenzene (TCB, bp  $196^\circ\text{C}$ ). Under these conditions, it took only five minutes to prepare  $\text{Zn}[(15\text{C}5)_4\text{Pc}]$  in 82% yield [50]. Starting from a solution of  $\text{Li}_2[(18\text{C}6)_4\text{Pc}]$  in DMSO and  $\text{Zn}(\text{OAc})_2$ , it was possible to increase yield of  $\text{Zn}[(18\text{C}6)_4\text{Pc}]$  up to 95% [51].

Direct synthesis was used to prepare  $\text{Ag}[(15\text{C}5)_4\text{Pc}]$  in 40% yield via the reaction between  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  and  $\text{AgNO}_3$  in dichloroethane/AcOH mixture in the presence of sodium acetate. The vanadyl complex was obtained in 30% yield via prolonged refluxing of  $\text{VOSO}_4$  and  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  in DMF [52].

Reaction of  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  with  $\text{SnCl}_4$  in quinoline yielded 35% of  $[(15\text{C}5)_4\text{Pc}]\text{SnCl}_2$  [38]. Interestingly, an introduction of the group IVa metals significantly increased the thermal stability of CRPc's in comparison with the parent ligand  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$ , while the introduction of *d* metals (Co, Ni, Fe) generally decreased the decomposition temperature [25].

To prepare aluminum crown-phthalocyaninates, both template and direct approaches were used. The

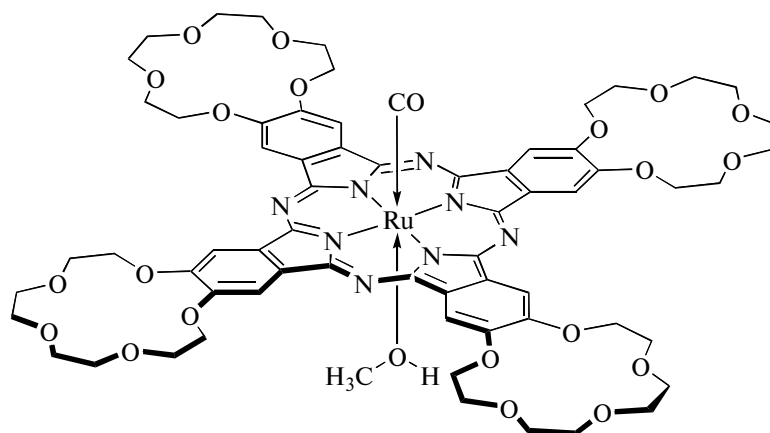
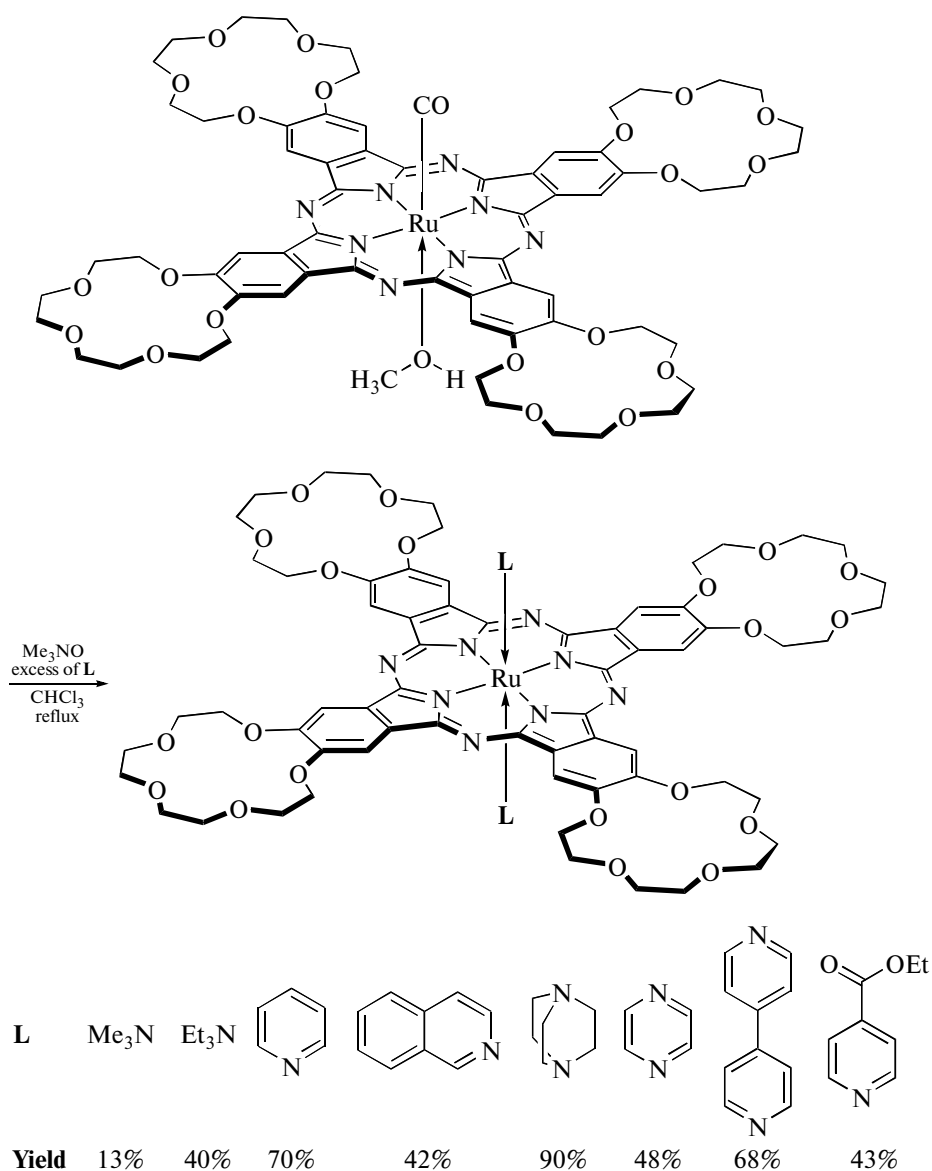


Fig. 4. Ruthenium crown-phthalocyaninate [(15C5)<sub>4</sub>Pc]Ru(CO)(CH<sub>3</sub>OH) [77].



Scheme 8. Decarbonylation of ruthenium tetra-15-crown-5-phthalocyaninate with simultaneous introduction of N-donor ligands [79–83].

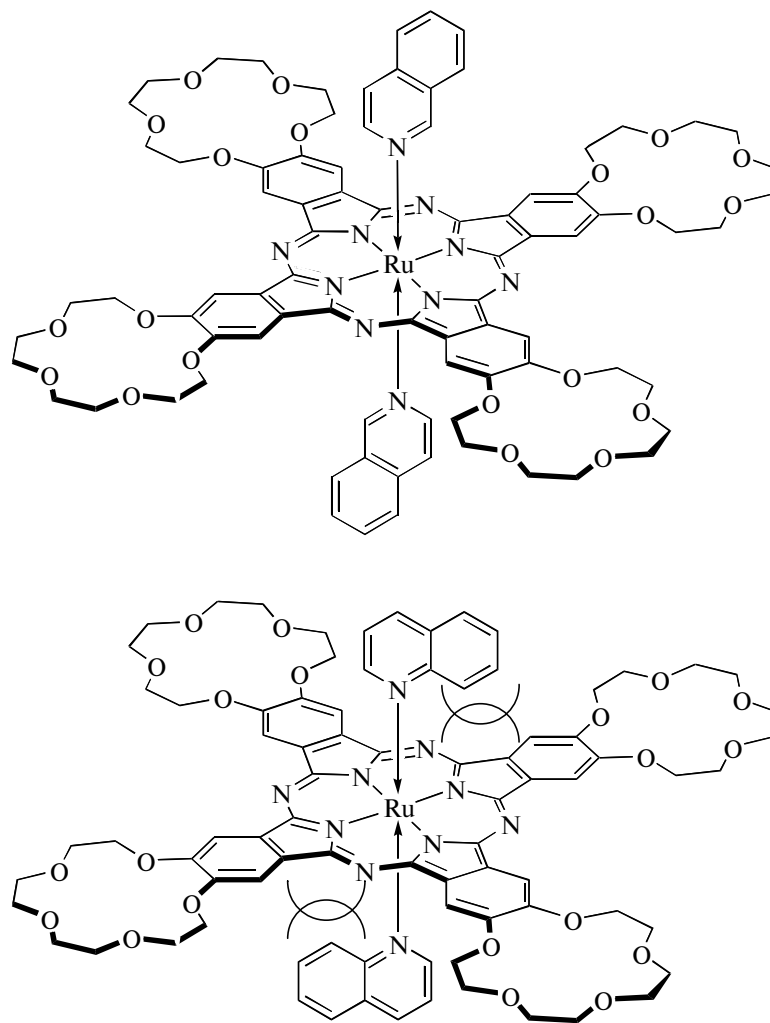


Fig. 5. Steric hindrance which precludes the formation of  $[(15C5)_4Pc]Ru(qnl)_2$  (bottom) [79].

condensation of DCB15C5 with  $Al_2(SO_4)_3$  in the presence of  $(NH_4)_2MoO_4$  yielded 54% of  $[(15C5)_4Pc]Al(OSO_3H)(H_2O)$  [53]. Direct metalation of  $H_2[(15C5)_4Pc]$  by aluminum ions could be achieved when this free base was treated with  $Al(acac)_3$  in a mixture of TCB and 1-octanol in the presence of phenol. It was proposed that the role of PhOH in this reaction involved the formation of  $Al(OPh)_3$  intermediate species, which was more active under the reaction conditions. After chromatographic purification, the complex  $[(15C5)_4Pc]Al(OH)$  was isolated in 96% yield [54].

A similar approach was used to prepare the gallium complex; it was obtained via the reaction of  $H_2[(15C5)_4Pc]$  with  $Ga(acac)_3$  and PhOH in a mixture of TCB and *i*-AmOH (yield, 97%) [55].

Indium complex could also be prepared by direct metalation of  $H_2[(15C5)_4Pc]$ . In this case, addition of phenol was not required, interaction of two reagents

occurred smoothly in refluxing 1-chloronaphthalene, yielding  $[(15C5)_4Pc]In(OH)$  in 90% yield after chromatographic purification [56].

Interest was drawn to Al(III), Ga(III), In(III) and Zn(III) CRPc's by their potential application as water-soluble agents for photodynamic therapy, due to their high energy and lifetime of the triplet state, resulting in the ability to generate singlet oxygen. Previously for this aim, a mixture of water-soluble sulfonated aluminum phthalocyaninates, known under the trade name "Photosense"— $[(SO_3Na)_nPc]Al(OH)$ ,  $n = 2-4$ , was prepared in Niopik (Organic Intermediates and Dyes Institute, Moscow, Russia) [57]. "Photosense" successfully passed clinical studies and its efficiency was demonstrated in the treatment of squamous cell skin cancer, breast, oral, lung and larynx cancers. Water soluble CRPc's were proposed as an alternative to "Photosense", since these compounds can be prepared as a

single isomer, providing consistency of chemical properties. In vivo tests of [(15C5)<sub>4</sub>Pc]Al(OSO<sub>3</sub>H)(H<sub>2</sub>O) showed that it had high affinity for lung, liver and spleen tumors. However, the compound turned out to be non-toxic upon light irradiation because of its high tendency to aggregation in aqueous media, thereby quenching luminescence and the PDT effect [58].

### 1.5. Synthesis of Ruthenium(II) Crown-Phthalocyaninates

The peculiarities of ruthenium coordination chemistry afford formation of different structural types of complexes with Por and Pc ligands [59]. These are complexes with different axial ligands [60–65], dimeric complexes with a Ru = Ru bond [60, 66–68],  $\mu$ -C [69] and  $\mu$ -N dimeric species [70],  $\mu$ -O polymers [71], and oligomers linked by different bridging ligands [61, 72, 73]. Unique photophysical properties of ruthenium complexes have resulted in wide applications of these compounds as components of photo- and electro-active materials and devices [59, 74–76]. In this respect, the interest in Ru(II) CRPc's is not surprising.

To prepare the first crown-substituted ruthenium phthalocyaninate, a template condensation of DCB15C5 was performed in the presence of various metal sources [77]. Fusion of DCB15C5 with different ruthenium complexes, namely RuCl<sub>3</sub> · 3H<sub>2</sub>O, Ru(DMSO)<sub>4</sub>Cl<sub>2</sub> or [Ru<sub>2</sub>(OAc)<sub>4</sub>Cl]<sub>n</sub> at 250°C for two to four hours yielded Ru(II) tetra-15-crown-5-phthalocyaninate [(15C5)<sub>4</sub>Pc]Ru(CO)(MeOH) regardless of the starting compound used (Fig. 4). This complex, containing CO and MeOH molecules as axial ligands, was isolated after chromatographic purification on alumina using CHCl<sub>3</sub>/MeOH gradient elution (apparently the methanol molecule was introduced at this stage). The presence of CO was confirmed by infrared spectroscopy (1934 cm<sup>-1</sup>—stretching vibrations of the carbonyl group), as well as by the ESI-MS spectrum (using MALDI TOF mass-spectrometry only [(15C5)<sub>4</sub>Pc]Ru<sup>+</sup> is observed together with the dimer, trimer and tetramer). The upfield-shifted resonance signal of an axially coordinated methanol molecule was observed in its NMR spectrum of the complex in CDCl<sub>3</sub> ( $\delta$  = -0.06 ppm).

The presence of CO in the isolated complex prepared from RuCl<sub>3</sub> · 3H<sub>2</sub>O, Ru(DMSO)<sub>4</sub>Cl<sub>2</sub> and [Ru<sub>2</sub>(OAc)<sub>4</sub>Cl]<sub>n</sub> was somewhat unexpected, since there was no direct carbonyl source in the system. It was supposed that the CO molecule was formed as a result of decomposition of DCB15C5 during high-temperature template synthesis by analogy with what occurs during preparation of unsubstituted ruthenium phthalocyaninates and RuCl<sub>3</sub> · 3H<sub>2</sub>O [78]. Decomposition of DCB15C5 explained the low yields in these reactions (10–15%). However, when ruthenium car-

bonyl Ru<sub>3</sub>(CO)<sub>12</sub> was used both as metal and CO source, the yield was increased to 80%.

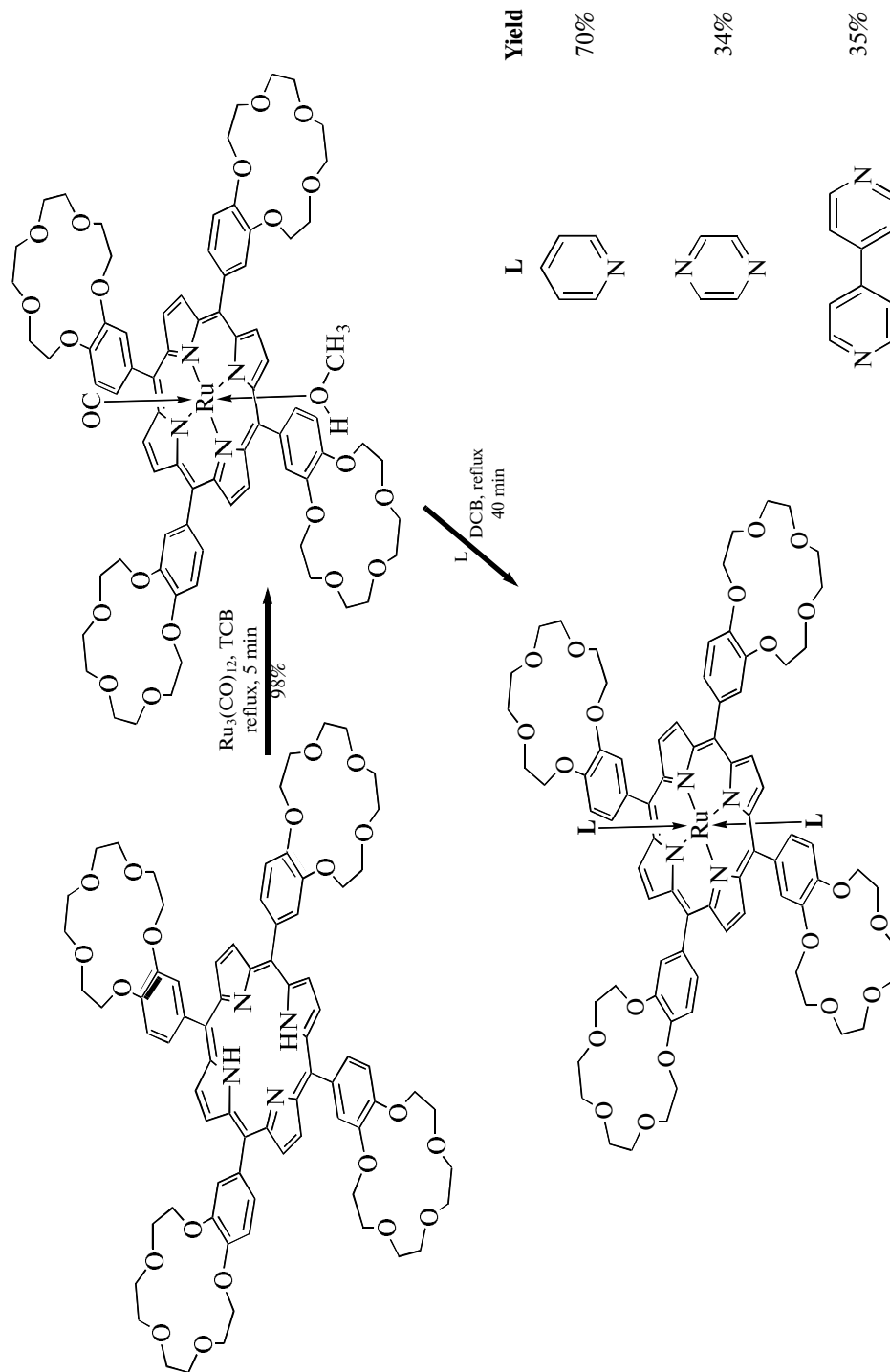
The presence of a carbonyl ligand strongly bound to ruthenium restricted the possibilities for further preparation of complexes with other axial ligands. To remove the CO molecule and to replace it with other donor ligands, photochemical decarbonylation was typically applied. However, this led to partial decomposition of the metal complex, resulting in low yields of target products. To overcome this obstacle, trimethylamine oxide was, for the first time, proposed as a decarbonylation agent for ruthenium phthalocyanines, using the example of [(15C5)<sub>4</sub>Pc]Ru(CO)(MeOH) [79]. In the presence of four equivalents of this reagent and excess of N-donor axial ligands, [(15C5)<sub>4</sub>Pc]Ru(CO)(MeOH) could be smoothly converted into [(15C5)<sub>4</sub>Pc]Ru(L)<sub>2</sub>, where L = amines or nitrogen-containing heterocycles (Scheme 8) [79–83]. The reaction can be easily monitored by UV-Vis spectroscopy since removal of CO and introduction of N-donor ligands is followed by a strong hypsochromic shift of the Q-band from 655 to 625–630 nm.

It is noteworthy that performing this reaction at room temperature resulted in formation of ~20% of a side complex, containing one molecule of trimethylamine, [(15C5)<sub>4</sub>Pc]Ru(L)(Me<sub>3</sub>N), which could be separated from the symmetrical compound by column chromatography. However, refluxing of the reaction mixture resulted in evaporation of volatile Me<sub>3</sub>N; only [(15C5)<sub>4</sub>Pc]Ru(L)<sub>2</sub> was obtained under these conditions. In the case of DABCO, pyrazine and bipyridyl, no formation of polymeric compounds was observed apparently because of the applied excess of N-donor ligand.

Application of commercially available quinoline to make bis-quinoline complexes resulted in formation of isoquinoline adducts. This was in line with the previously reported results—1–2% of isoquinoline was found in commercially available quinoline by GC/MS [84]. When pure quinoline was applied, no more complex formation was observed because of steric hindrance, which prevents the introduction of quinoline into ruthenium coordination sphere (Fig. 5).

Ruthenium CRPc's with axial N-donor ligands were found to be prominent photosensitizers for polymeric photorefractive composites [112, 115–122].

The applicability of Me<sub>3</sub>NO in preparation of ruthenium tetrapyrrolic complexes with N-donor axial ligands was demonstrated on the example of tetra-*meso*-(benzo-15-crown-5)-substituted porphyrinate [(*meso*-B15C5)<sub>4</sub>Por]Ru(CO)(MeOH), prepared by direct metalation of corresponding free-base porphyrin with Ru<sub>3</sub>(CO)<sub>12</sub> in refluxing TCB (Scheme 9) [89]. In contrast to crown-substituted phthalocyanines, the decarbonylation of ruthenium porphyrinate required harsh conditions—refluxing in dichlorobenzene (DCB, bp 180°C).



Scheme 9. Synthesis of ruthenium(II) crown-porphyrinates [89].

## 2. SYNTHESIS AND PROPERTIES OF RARE-EARTH ELEMENTS CROWN-SUBSTITUTED PHTHALOCYANINES

The ability to form, along with monophthalocyanines, stacking coordination oligomers with metal-ligand compositions 1 : 2 and 2 : 3 (the so-called double- and triple-decker complexes, respectively), is the characteristic feature of coordination chemistry of rare earth elements (lanthanides, Y, Sc) with Pc ligands [90, 91]. Every structural type of these complexes reversibly forms several redox-forms. Moreover, each of them may, in turn, participate in acid-base equilibria [92–99], making them prominent electroactive materials (electrochromic devices and sensors [100, 101], etc.). The redox-potentials, as well as maxima of absorption bands in UV-Vis-NIR spectra of sandwich complexes, reveal linear dependencies vs. the ionic radii of lanthanides because of the effects associated with the intramolecular  $\pi$ – $\pi$  interaction between stacked ligands [102–104].

Once the crown-ether groups are attached to the periphery of double-decker and triple-decker REE complexes, their properties can be tuned by molecular recognition of metal cations [105]. From this viewpoint, the elaboration of efficient and regioselective synthetic approaches towards crown-substituted REE complexes and their further thorough characterization (certainly including studies of cation-induced assemblies) becomes an important many-sided task.

### 2.1. Homoleptic Complexes

The story of homoleptic sandwich CRPc's of REE investigation started in 1989 when  $\text{Lu}[(15\text{C}5)_4\text{Pc}]_2$  was first synthesized [106]. The aim was to make redox-active electrochromic complex with three redox-states ( $\text{red}^+$ ,  $\text{green}^0$  and  $\text{blue}^-$ ) accessible in water solutions. It was anticipated that the presence of crown-ether groups would afford better penetration of counterions into the films formed by the reduced (blue) form of the complex to improve their stability. To make  $\text{Lu}[(15\text{C}5)_4\text{Pc}]_2$ , DBU-assisted template condensation of DCB15C5 in the presence of  $\text{Lu}(\text{OAc})_3$  was performed in refluxing 1-hexanol, yielding 19–22% of the required complex [106, 107]. In the case of DCB18C6, the corresponding double-decker complex was obtained in 7% yield [107].

Unsubstituted phthalonitrile is known to react with REE acetates at high temperature, yielding sandwich phthalocyaninates [108, 109]. This approach was applicable in the case of DCB15C5, reacting with  $\text{Lu}(\text{OAc})_3$  in a melt (reagent ratio 8 : 1, 265°C, 40 min) to yield the green double-decker complex  $\text{Lu}[(15\text{C}5)_4\text{Pc}]_2$  (35%) [110]. The formation of this complex was reliably confirmed by its characteristic absorption in the NIR (1390 nm) and also by its singlet signal in the ESR spectrum ( $g = 2.0029$ ). This data supports the presence of an

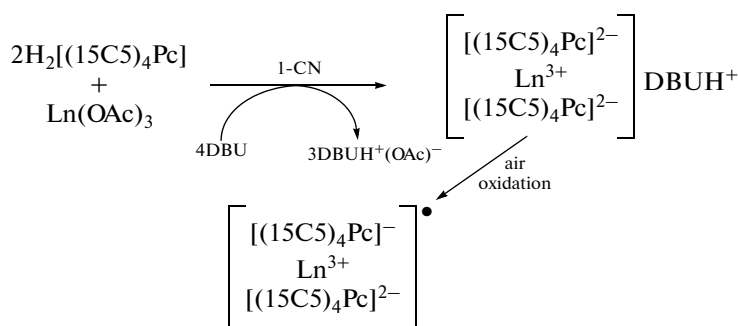
unpaired electron, delocalized over two CRPc ligands:  $[(15\text{C}5)_4\text{Pc}^{2-}]\text{Lu}^{3+}[(15\text{C}5)_4\text{Pc}^{\bullet-}]$ . Together with the green complex, there was also isolated a diamagnetic blue complex. Interestingly, a minor amount of brown-colored compound was isolated from the reaction mixture; its MALDI TOF spectrum corresponded to the Lu(III) double-decker complex. This compound was attributed to the Lu(III) double-decker complex with covalently bonded ligands [111] because of the similarity of its UV-vis spectrum ( $\lambda_{\text{max}}$  424 and 475 nm) with the spectrum of  $\text{Ti}(\text{Pc})_2$ , whose stapled structure was confirmed by X-ray diffraction analysis.

The structure of the blue complexes, formed in template synthesis of  $\text{Ln}(\text{Pc})_2$  was an object of a long-lasting discussion [96, 91]. Several opinions were expressed about its structure: a protonated double-decker complex  $[\text{H}^+(\text{Pc}^{2-})\text{Ln}^{3+}(\text{Pc}^{2-})]^0$ , a double-decker complex of a tetravalent lanthanide  $[\text{Ln}^{4+}(\text{Pc}^{2-})_2]^0$  or a triple-decker complex  $[(\text{Ln}^{3+})_2(\text{Pc}^{2-})_3]^0$ . This uncertainty originated from the similarity of UV-Vis spectra of all these species, since it is mainly determined by transitions in the phthalocyaninato-dianions ( $\text{Pc}^{2-}$ ). This discussion ended with the example of the “blue form”, isolated from the reaction between DCB15C5 and  $\text{Lu}(\text{OAc})_3$ , when its single-crystal X-ray analysis was performed, giving the *first* structural evidence of a dilanthanide trisphthalocyaninate [112].

Using the template reaction of DCB15C5 with  $\text{Yb}(\text{OAc})_3$ , resulted in a decrease of the yield of sandwich complexes; only traces of the middle lanthanides complexes were obtained. In order to increase the yields, template synthesis of double-decker complexes can be performed in solution. For example,  $\text{Eu}[(12\text{C}4)_4\text{Pc}]_2$  and  $\text{Eu}[(18\text{C}6)_4\text{Pc}]_2$  could be prepared in ~30% yield via refluxing of the corresponding DCB-crown-ether,  $\text{Eu}(\text{acac})_3$ , and DBU in pentanol [29]. Similarly,  $\text{Tb}[(15\text{C}5)_4\text{Pc}]_2$  was prepared in excellent yield (79%) by reaction of DCB15C5 and  $\text{Tb}(\text{acac})_3$  in refluxing isoamyl alcohol and DBU [113]. However, under these conditions triple-decker complexes could not be obtained.

Much attention was therefore paid to studies of direct metalation of metal free CRPc's with REE acetates in high boiling solvents in the presence of a base, which assists in removal of protons from the phthalocyanine cavity [114–116]. All of these factors (nature of the solvent used for the reaction, nature of the REE ion, its counterion and the presence of the base) strongly influence the pathway of metalation, making it a selective approach towards obtaining complexes with different structures.

The reaction of  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  with  $\text{Lu}(\text{OAc})_3$  in refluxing *o*-dichlorobenzene (DCB, b.p. 180°C) in the presence of a base, DBU or Phen (molar ratio 1 : 3 : 40), was found to quantitatively yield lutetium monophthalocyaninate  $[(15\text{C}5)_4\text{Pc}]\text{Lu}(\text{OAc})(\text{base})$  [114]. The



**Scheme 10.** Direct metalation of tetra-15-crown-5-phthalocyanine,  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  with lanthanide acetates and the role of DBU in the stabilization of the intermediate anionic form of the double-decker complex.

composition of the coordination sphere of the lutetium ion was confirmed by NMR studies. These water soluble complexes were found to be promising agents for photodynamic therapy due to their luminescent properties [117]. In the same manner, monophthalocyaninates of other REE complexes (Sm, Tb, Dy, Er, Tm, Y) were later prepared [118–121].

Further investigations of this reaction revealed that an increase in the reaction temperature and use of refluxing 1-chloronaphthalene (1-CN, bp 260°C) as a solvent, led to the formation of sandwich complexes. The reactions of  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  with  $\text{Ln}(\text{OAc})_3$ ,  $\text{Ln} = \text{Gd}, \text{Yb}, \text{Lu}$  and DBU (reagent ratio 1.5 : 1 : 15) yielded 15–20% of double-decker and 10–15% of triple-decker complexes [115].

Special interest in triple-decker REE complexes can be explained by their characteristic features, arising from their chemical structure, namely:

- the presence of three stacked Pc ligands in the molecule results in a rich redox-chemistry, providing compounds with several reversible well-resolved redox-transitions over a relatively narrow range of potentials [104, 122]. Due to this advantage, the prototypes of molecular information storage devices were developed on the basis of triple-decker REE complexes with tetrapyrrole ligands [123].

- close proximity of paramagnetic lanthanide ions (3.3–3.5 Å) in these molecules results in interaction between their *f* shells, resulting in interesting magnetic properties [124, 125].

The low yields of triple-decker complexes by both template and direct synthesis of REE CRPc's inspired the search for the optimal conditions needed in regioselective synthesis of these compounds. A solution to this problem comes from an analysis of the mechanism for formation of the triple-decker complexes by direct metalation.

A spectrophotometric monitoring of the reaction of  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  with  $\text{Ln}(\text{OAc})_3$  and DBU in refluxing 1-CN revealed that the first step in this reaction is the formation of a monophthalocyaninate, which subsequently forms the double-decker and triple-decker

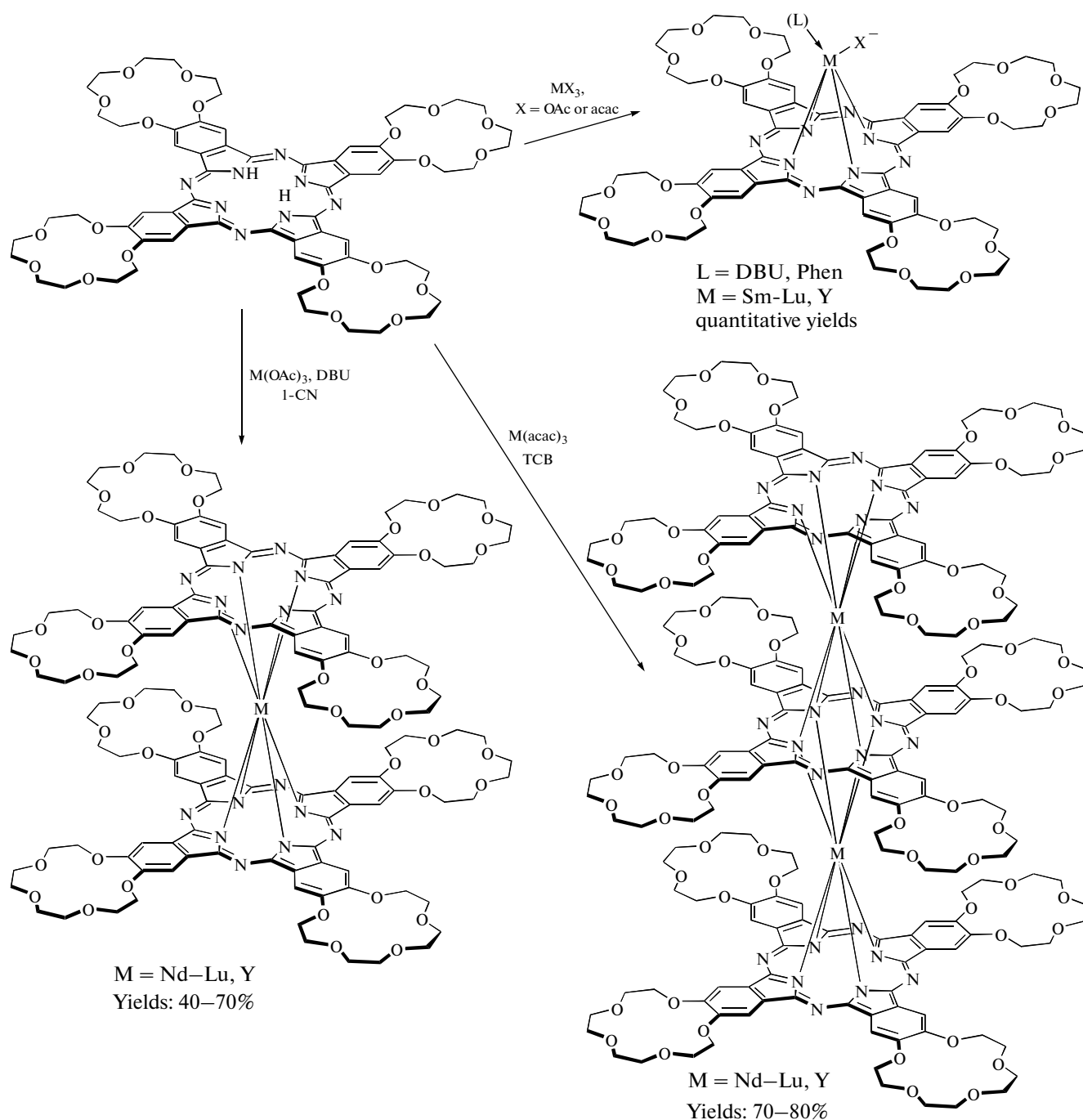
complexes. Apparently, the formation of triple-decker complexes is stepwise as well. Thus, the yield of triple-decker should depend on the activities of the intermediates; namely, mono- and double-deckers, which in turn depend on the stabilizing factors.

In the case of monophthalocyaninate, the stabilization is associated with a with the saturation of the lanthanide coordination sphere with N-donor DBU molecules; a formation of  $[(15\text{C}5)_4\text{Pc}]\text{Lu}(\text{OAc})(\text{DBU})$  was evidenced from NMR studies [114].

In the case of the double-decker complex, its stabilization also arises from the presence of DBU (Scheme 10). Its protonated form acts as a counterion of an anionic double-decker complex, formed under metalation conditions. Upon isolation of this anionic complex, its air oxidation leads to the formation of the neutral form  $[(15\text{C}5)_4\text{Pc}^{2-}]\text{Ln}^{3+}[(15\text{C}5)_4\text{Pc}^-]$ .

Therefore, the idea came to exclude DBU from this reaction. However, the base was still needed to remove protons from the  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  molecule. The solution was to use REE acetylacetonates instead of acetates, since acetylacetonate is a weak CH-acid ( $\text{pK}_a$  9) and its anion should be a strong conjugated base. The reaction between  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  and  $\text{M}(\text{acac})_3$  was studied ( $\text{M} = \text{Pr}, \text{Nd}, \text{Tb}, \text{Er}, \text{Lu}, \text{Y}$ ) in the absence of DBU. Performing this reaction in refluxing 1-chloronaphthalene (bp 260°C) or 1,2,4-trichlorobenzene (bp 216°C) resulted in formation of the triple-decker complexes  $\text{M}_2[(15\text{C}5)_4\text{Pc}]_3$  in high yields (70–80%) [118, 119, 121, 126]. The double-decker complexes were isolated only in trace amounts. Once DBU was added to this reaction, the yield of triple-decker complexes was decreased to 40–50%, because of the increase in yield of the double-decker complexes. Replacing acetate with acetylacetonate in the synthesis of monophthalocyaninates in DCB does not influence the pathway of the reaction.

The overall scheme of direct metalation of  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  with middle and late Ln salts (as well as Y) is given in Scheme 11.



**Scheme 11.** Direct metalation of  $H_2[(15C5)_4Pc]$ , targeted to regioselective synthesis of REE crown-substituted mono-, bis- and trisphthalocyaninates.

The developed synthetic approaches are applicable for preparation of CRPc complexes of middle and late lanthanides together with Y. However, some problems are evident when extremes in the REE family are taken, namely those with very large La(III) and very small Sc(III) ions.

When  $La(OAc)_3$  was used in common conditions for the synthesis of monophthalocyaninates, no com-

plex formation was observed [127]. Increasing the amounts of the added DBU and metal salt shifted the equilibrium. The expected monophthalocyaninate was not obtained and instead, the anionic form of double-decker complex, stabilized with DBU was isolated,  $La^{3+}[(15C5)_4Pc^{2-}]_2 \cdot DBUH^+$ . The stability of this complex originated from the more positive oxidation potential of La(III) anionic double-decker com-



plexes among complexes of other metals, as evidenced by electrochemical studies of  $\text{Li}^+[\text{Ln}(\text{Pc})_2]^-$ ,  $\text{Ln} = \text{La} - \text{Lu}$  except Ce and Pm [98].

Once 1-CN was used for this reaction, the amounts of  $\text{La}(\text{OAc})_3$  and DBU needed for the complex formation could be decreased. Chromatographic separation of the reaction products also afforded the anionic double-decker complex, accompanied by neutral  $\text{La}[(15\text{C}5)_4\text{Pc}]_2^0$ . Spectrophotometric titration with a solution of DBU resulted in reduction of the complex, thus justifying the role of DBU in stabilizing anionic forms of the double-decker complexes.

To prepare the lanthanum triple-decker complex, the reaction of  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  with  $\text{La}(\text{acac})_3 \cdot n\text{H}_2\text{O}$  was performed. Both UV-Vis and MALDI TOF MS spectra showed formation of the target compound  $\text{La}_2[(15\text{C}5)_4\text{Pc}]_3$ . However, attempts to isolate these species by column chromatography on alumina resulted in decomposition with formation of the double-decker complex and  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  [128]. Separation of the reaction products by more gentle gel-permeation chromatography on Bio-Beads SX1 yielded a fraction containing the target triple-decker complex. According to UV-vis spectroscopy, this compound gradually decomposed with the formation of a mixture of  $\text{La}[(15\text{C}5)_4\text{Pc}]$  and  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$ .

Scandium, having the smallest ion among REE, also shows a peculiar behavior in the formation of CRPc complexes [129]. It does not form triple-decker complexes, probably because of steric hindrance, but the reaction of  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  with  $\text{Sc}(\text{acac})_3$  yields the double-decker complex  $\text{Sc}[(15\text{C}5)_4\text{Pc}]_2$  in 70% yield after refluxing in TCB/1-octanol (7 : 1 vol). In the absence of 1-octanol, the complex formation did not occur. The conditions required to obtain the monophthalocyaninate are harsher than those for the rest of REE. Reaction of  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  was performed in TCB in the presence of imidazole as a base, yielding 97%  $[(15\text{C}5)_4\text{Pc}]\text{Sc}(\text{OAc})$  after column chromatography on alumina (elution with a  $\text{CHCl}_3$ -MeOH-AcOH mixture) [129].

Among lanthanides, cerium stands apart due to its ability to exist in both tri- and quadrivalent oxidation states in complexes with tetrapyrrolic ligands, the nature of which predetermines the exact oxidation state of the cerium ion [130–134]. An equilibrium between Ce(III) and Ce(IV) oxidation states is also possible in solutions of some heteroleptic porphyrinates and naphthalocyaninate [132, 133].

The above described direct metalation of  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  was used to prepare cerium CRPc complexes [135]. With this intention, Ce(III) salts (acetate or acetylacetonate) were refluxed with the ligand in the presence of DBU in 1-CN. Use of solvents with a lower b.p. did not result in complex formation. After chromatographic purification the double-decker complex was isolated in 60% yield. The

NMR spectrum of the complex exhibited resonances in the usual range for diamagnetic complexes, suggesting the quadrivalent state of the cerium ion,  $\text{Ce}^{4+}[(15\text{C}5)_4\text{Pc}^{2-}]_2$ . Additional proof of the Ce oxidation state was obtained by analysis of UV-Vis spectrum in comparison with UV-Vis spectra of double-decker complexes of other rare earth elements, which can form complexes only in trivalent state [135].

X-ray diffraction analysis was performed for two isostructural complexes  $\text{Ln}[(15\text{C}5)_4\text{Pc}]_2$ ,  $\text{Ln} = \text{Ce}$  [136] and Yb [137]. It was found that the Ce–N bond length [2.41(3)–2.43(3) Å] is shorter than that of Yb–N [2.254(11)–2.410(13) Å]. This shortening suggests that the complex is formed by Ce(IV), since its ionic radius (0.970 Å) is smaller than Yb(III) (0.985 Å), while Ce(III) has a larger radius (1.143 Å).

Due to reversible formation of colored redox-forms, metal CRPc's are used as an electrochromic component of sensors for redox-active gases [138–143]. For this purpose, cast films are typically applied. However, better functionality can be achieved if Langmuir–Blodgett technology is applied, because it affords formation of films with a controlled thickness and morphology [144]. For example, optoelectronic sensors for tobacco smoke were prepared containing two types of films of Lu double-decker complexes, either transferred from a pure water surface ("green" film) or from a 0.01 M HCl solution subphase ("red" film) [101]. These green and red monolayers were able to distinguish between the main and the secondary streams, respectively, and exhibited higher sensitivity to increasing amounts of smoke than other bisphthalocyanines.

In the case of double-decker and triple-decker CRPc's, the presence of crown-ethers in these receptors increases their stability and decreases the oxidation potential of the molecules. Moreover, binding of alkali-metal ions by crown-ether groups can be used for tuning of electrophysical characteristics of these films [101, 106, 145].

The particular ability of the cerium ion to have different valent states in complexes with tetrapyrrole ligands can also be utilized in multistable optoelectronic devices. ITO electrodes, covered with Langmuir–Blodgett films of  $\text{Ce}[(15\text{C}5)_4\text{Pc}]_2$  were prepared and their electrochemical measurements were performed in the range of potentials from –0.2 V to 1.0 V [146]. It was demonstrated that this complex exhibits three redox transitions in this range, similar to the triple-decker complexes, while double-decker complexes with other REE(III) have only two redox transitions. Detailed analysis of these redox-processes revealed that a process at 415 mV corresponds to metal-centered redox transition in the couple  $[\text{Pc}^{2-}\text{Ce}^{3+}\text{Pc}^{\bullet}]^0/[\text{Pc}^{2-}\text{Ce}^{4+}\text{Pc}^{\bullet}]^+$ .

A change in the metal center oxidation state should lead to a change in the metal ionic radius, and as a consequence, to a change of distance between the

macrocycles. In the case of a large number of molecules in a stack, the change in this distance may lead to ca. 15% difference in linear dimensions of the supramolecular assemblies. This effect can be used for the development and creation of supramolecular devices capable of efficient conversion of chemical energy into mechanical work [147].

## 2.2. Heteroleptic Sandwich Complexes

Synthesis of heteroleptic double- and triple-decker complexes with tetrapyrrole ligands is one of the generally accepted approaches towards compounds with required physical-chemical properties such as UV-Vis-NIR absorption patterns, sets of redox-transitions, magnetic properties, etc. [148–150]. Combining crown-substituted tetrapyrrole ligands in sandwich complexes affords in-parallel tuning of ionophoric properties, giving a pathway to molecular blocks of well-defined supramolecular assemblies.

To prepare the first examples of heteroleptic crown-phthalocyaninates, the interaction of  $\text{Lu}(\text{Pc})(\text{OAc}) \cdot 2\text{H}_2\text{O}$  with  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  in refluxing 1-CN was performed. The reagents, taken in an equimolar ratio, yielded the double-decker complex  $[(15\text{C}5)_4\text{Pc}]\text{Lu}(\text{Pc})$  [151]. The ratio 3 : 1 produced the triple-decker complex  $(\text{Pc})\text{Lu}[(15\text{C}5)_4\text{Pc}]\text{Lu}(\text{Pc})$  [152], which was, in addition, the first reported heteroleptic triple-decker complex constructed of phthalocyanine ligands. Introduction of crown-ethers into the later complex resulted in a significant improvement in solubility in comparison with  $\text{Lu}_2(\text{Pc})_3$ , which was required for further spectroscopic studies. [153] Alternatively,  $[(15\text{C}5)_4\text{Pc}]\text{Lu}(\text{Pc})$  was prepared in 14% yield by fusion of  $\text{Lu}(\text{Pc})(\text{OAc}) \cdot 2\text{H}_2\text{O}$  with DCB15C5 (molar ratio 1 : 5) at 260°C for 30 minutes [154].

Both of these reactions are representatives of the so-called “raise by one storey” approach, wherein the ligands are introduced one by one, resulting in sandwich molecule. In the examples given above, previously synthesized macrocycles are taken for this purpose. This method is commonly used to prepare heteroleptic triple-decker complexes. For double-decker complexes another approach is typically applied, namely, template cyclotetramerization of phthalonitriles. The role of template is served by the metal ion, previously introduced into the monophthalocyaninate ligand.

Treatment of  $[(15\text{C}5)_4\text{Pc}]\text{Lu}(\text{OAc})$  with phthalonitriles bearing additional functional groups and DBU in high-boiling alcohols was used to prepare the complexes  $[(15\text{C}5)_4\text{Pc}]\text{Lu}[(\text{R})_8\text{Pc}]$  (Scheme 12). Among them, the complex bearing four chains of 6-oxyhecanoic acid ester. Terminal ester groups were then hydrolyzed and the obtained acid was grafted onto the silica surface [155, 156]. The series of  $[(15\text{C}5)_4\text{Pc}]\text{Ln}(\text{Pc})$  compounds, where  $\text{Ln} = \text{Sm}, \text{Dy}$  and  $\text{Tm}$  was prepared via the reaction of

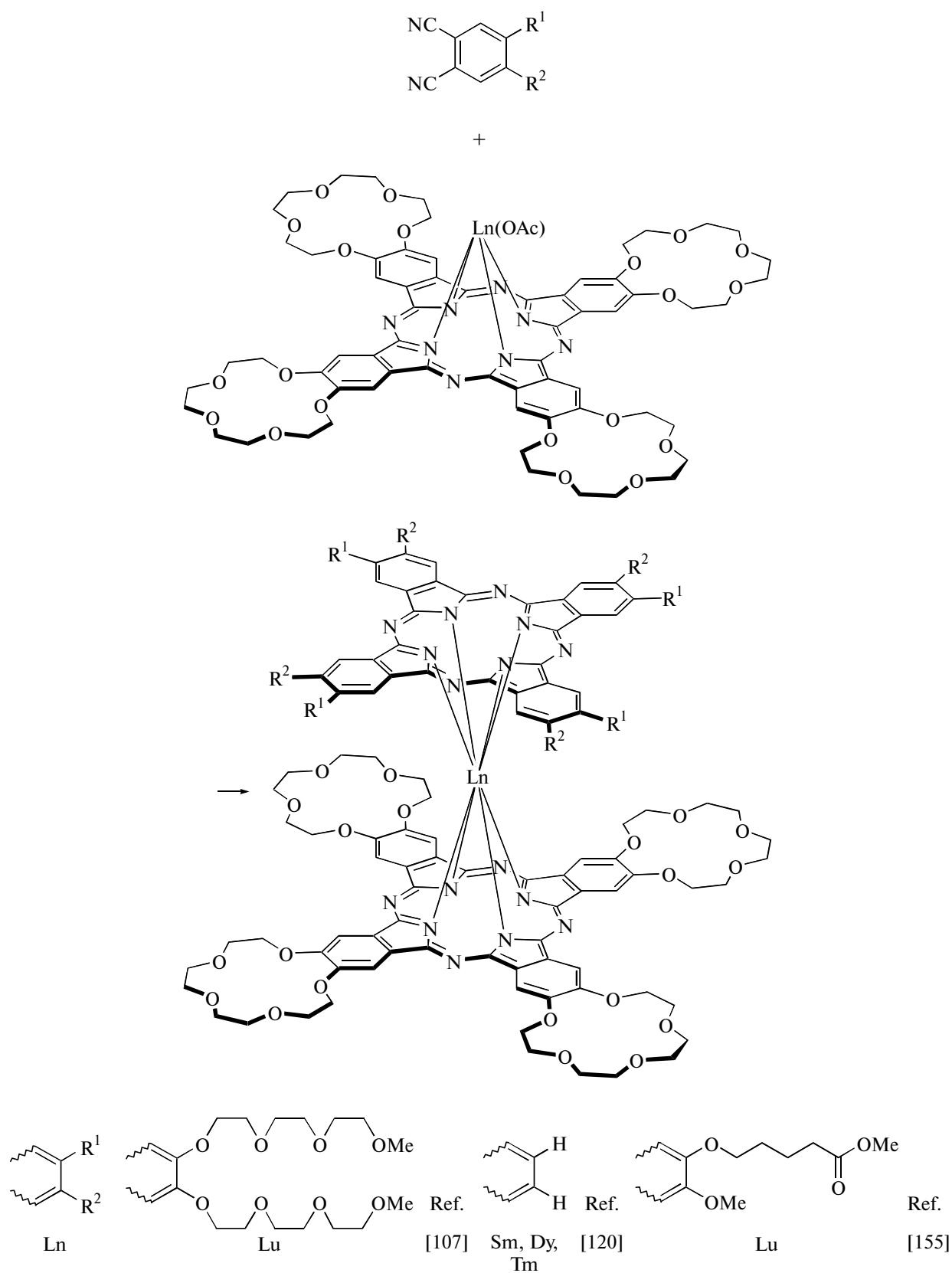
$[(15\text{C}5)_4\text{Pc}]\text{Ln}(\text{OAc})$  with unsubstituted phthalonitrile and DBU in refluxing 1-BuOH/1-CN [120].

Together with heteroleptic double-decker sandwiches, constructed of symmetric phthalocyanines, several examples of complexes containing low-symmetry ligands were reported. Among these was  $[(15\text{C}5)\text{Pc}]\text{Lu}(\text{Pc})$ , prepared via the reaction of DCB15C5 with excess of phthalonitrile, lutetium acetate and DBU in 1-hexanol [157]. Together with the target complex, unsubstituted and poly-crown substituted double-decker complexes were obtained. Therefore, multistep chromatographic purification was required. The yield of the isolated  $[(15\text{C}5)\text{Pc}]\text{Lu}(\text{Pc})$  was 3%. To decrease the number of side-products in the synthesis of heteroleptic double-decker complexes with low-symmetry phthalocyanines, two different approaches can be applied. One is the synthesis of low-symmetry phthalocyanine in the first step. It is then converted into the monophthalocyaninate which is treated with an appropriate phthalonitrile. In this manner, Lu and Dy double-decker complex with seven 15-crown-5 units and one hexanoic acid chain were prepared in 21% yield via the reaction of unsymmetrical monophthalocyaninate with DCB15C5 and DBU in refluxing 1-hexanol (Scheme 13) [47, 48]. Synthesis of the corresponding low-symmetry ligand is shown in Scheme 7. To prepare monophthalocyaninate, the low-symmetry ligand was treated with BuLi and then with corresponding  $\text{Ln}(\text{OAc})_3$ .

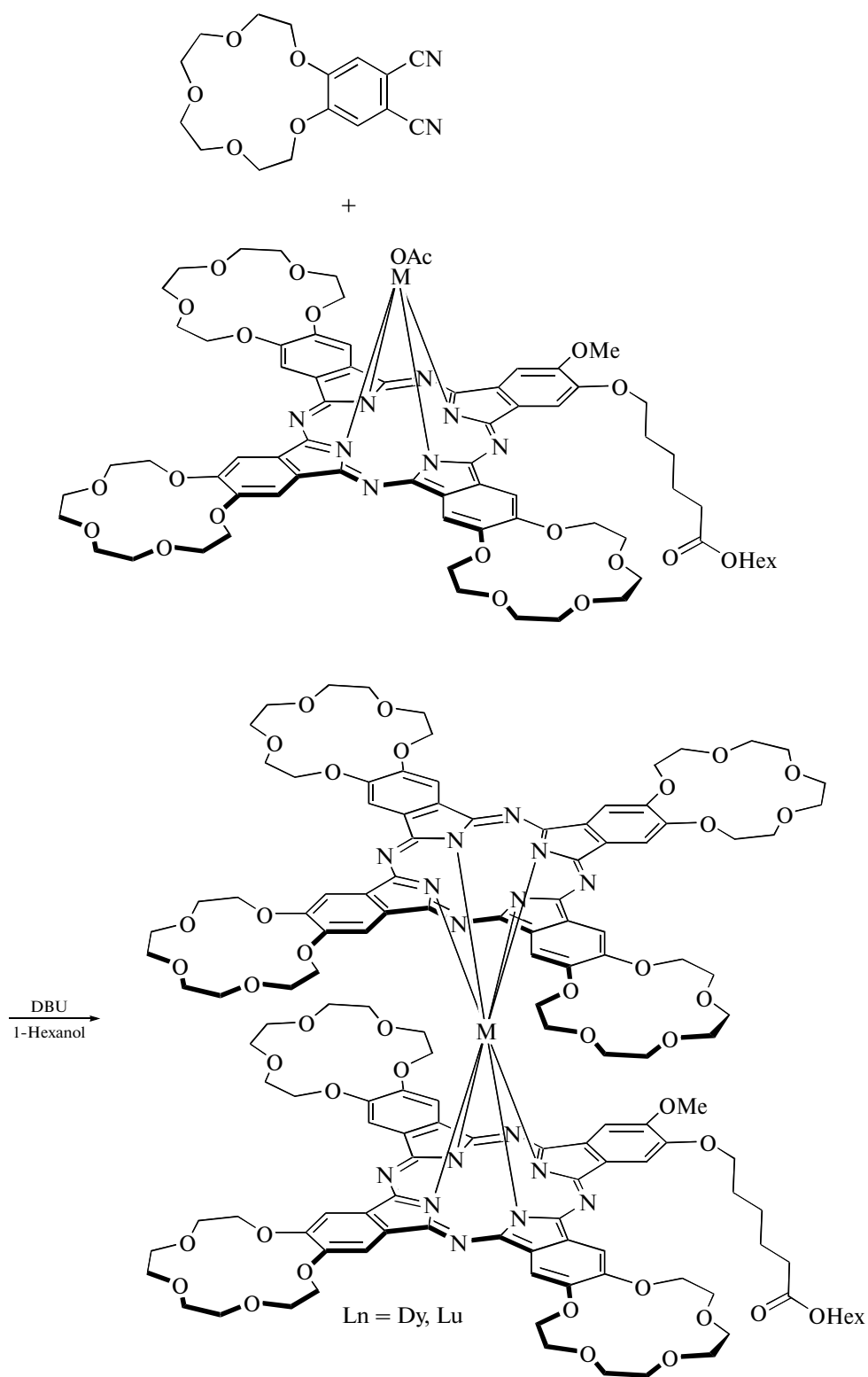
The series of heteroleptic double-decker europium complexes  $[(15\text{C}5)_n]\text{Eu}(\text{Pc})$ ,  $n = 1-4$  (including *opp*- and *adj*-isomers for  $n = 2$ ) were prepared by template cross-condensation of phthalonitrile and DCB15C5, performed in the presence of  $(\text{Pc})\text{Eu}(\text{OAc})$  and DBU in 1-pentanol [158] (Fig. 6). The mixture of heteroleptic complexes was separated by repetitive column chromatography on  $\text{SiO}_2$ , using gradient elution with a  $\text{CHCl}_3/\text{MeOH}$  mixture. The X-ray structure of  $[(15\text{C}5)_4\text{Pc}]\text{Eu}(\text{Pc})$  was determined. Comparative studies of the obtained complexes revealed a stepwise red-shift of the Q-band with increase in the number of the introduced crown-ether rings.

The synthesized complexes with low-symmetry ligands were useful for the synthesis of low-symmetry copper phthalocyaninates. In the presence of  $\text{Cu}(\text{OAc})_2$  in refluxing DMF  $[(15\text{C}5)_n]\text{Eu}(\text{Pc})$  complexes with  $n = 2$  and 3 were converted into the corresponding  $\text{Cu}[(15\text{C}5)_n\text{Pc}]$  complexes and isolated in 53–61% yield [159]. Similarly, low-symmetry Zn CRPc's were prepared and spectral characteristics of the obtained complexes were compared with TDDFT calculation results [160].

The described methods for preparation of heteroleptic double-decker complexes implies the synthesis of intermediate monophthalocyaninates. Therefore, their applicability is limited to complexes of middle and late lanthanides, while early lanthanides, which



**Scheme 12.** Synthesis of heteroleptic double-decker crown-phthalocyaninates.



**Scheme 13.** Synthesis of heteroleptic double-decker complexes bearing seven crown-ether units and one hexanoic acid chain [47, 48].

do not form stable monophthalocyaninates (La–Nd), are not applicable.

To solve this problem, an original approach was developed to prepare the first heteroleptic lanthanum

bisphthalocyaninate,  $[(15C5)_4Pc]La(Pc)$  [120]. This approach was based on the low stability of triple-decker lanthanum phthalocyaninates [128]. The idea was to prepare an intermediate heteroleptic triple-

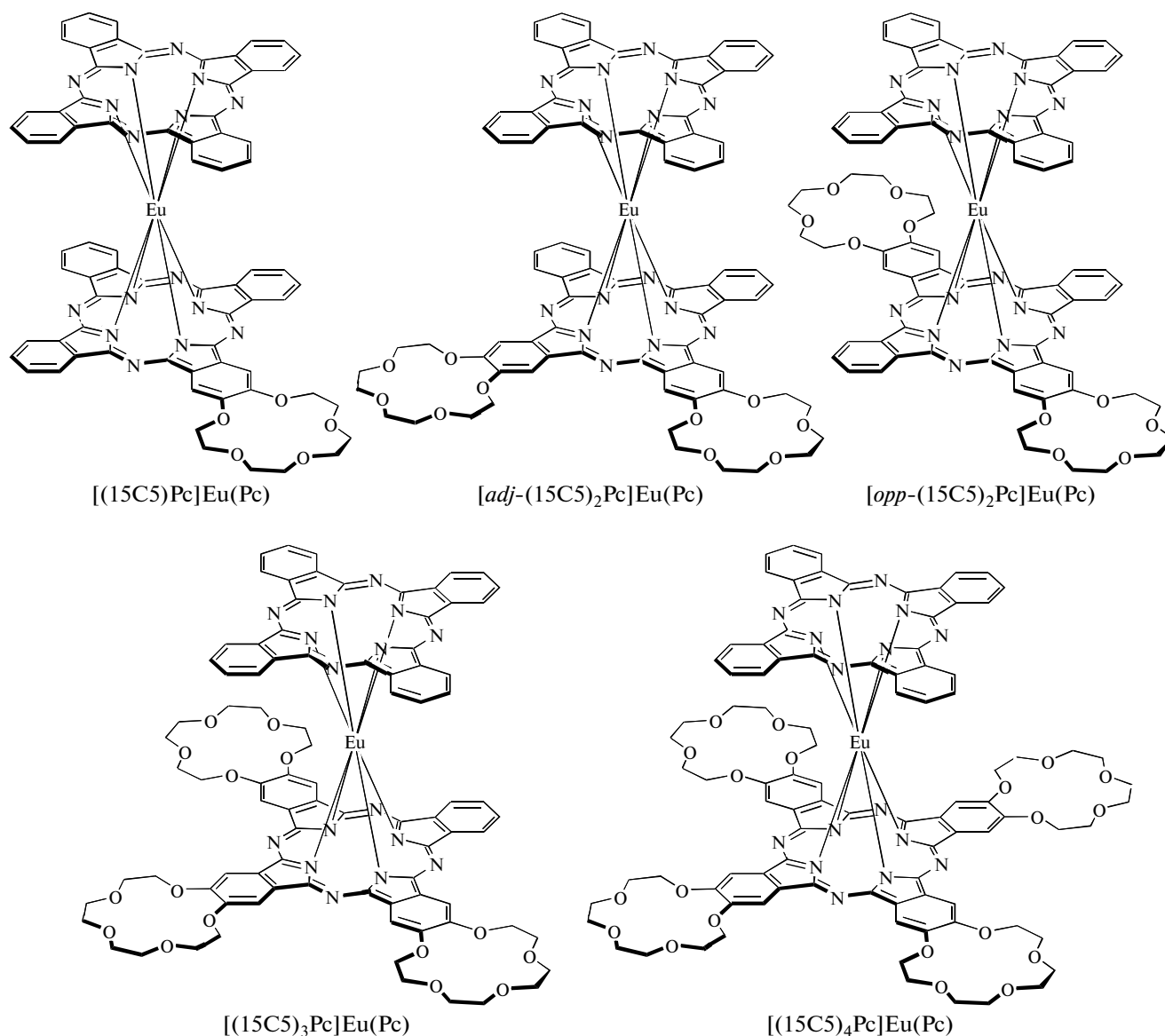
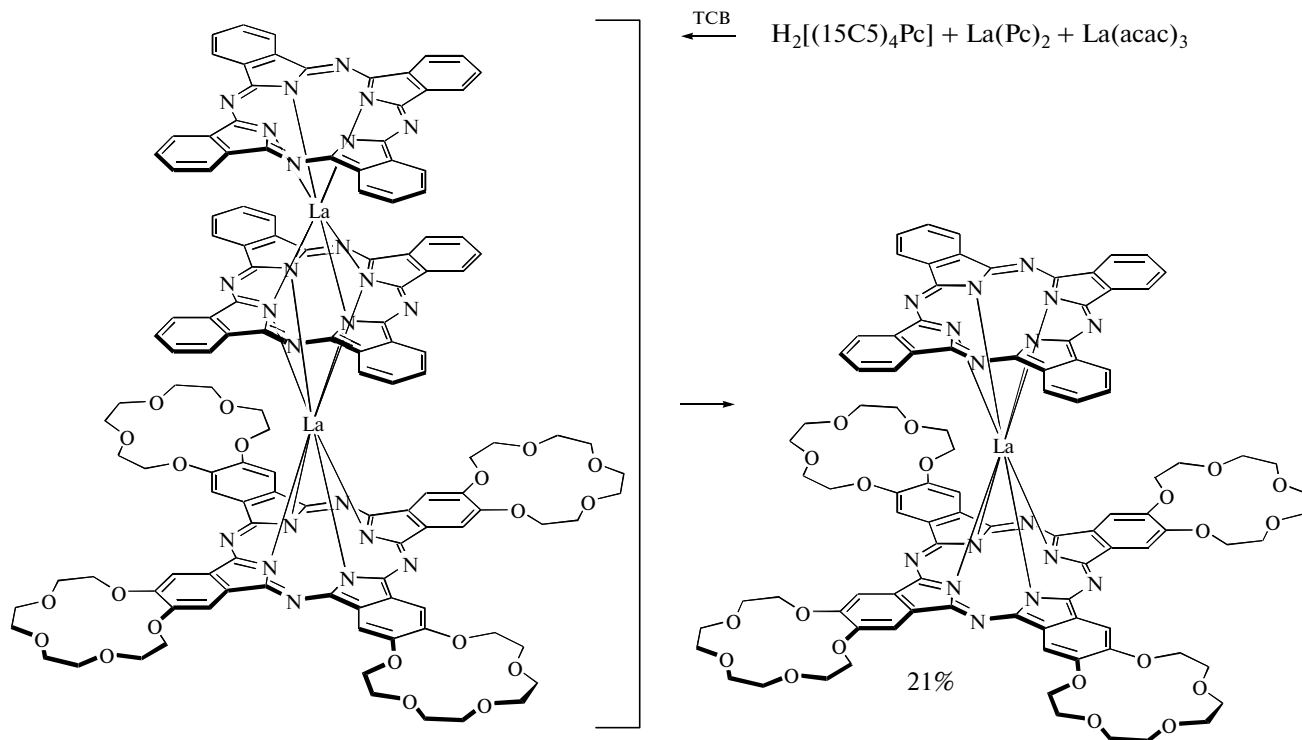


Fig. 6. Heteroleptic europium complexes  $[(15C5)_nPc]Eu(Pc)$ ,  $n = 1-4$  [158].

decker complex,  $[(15C5)_4Pc]La(Pc)La(Pc)$ , which would further decompose with formation of the target double-decker complex. With this aim the reaction of  $La(Pc)_2$  with  $H_2[(15C5)_4Pc]$  and  $La(acac)_3$  was performed in refluxing trichlorobenzene (Scheme 14). The high tendency of lanthanum complexes to form polar anionic forms [127] complicated chromatographic separation of reaction products. Therefore, a modified sorbent, alumina coated with  $MnO_2$ , was proposed to avoid formation of anionic forms. To prepare this sorbent, alumina was added to a solution of  $KMnO_4$  in acetone, and methanol was added to reduce  $Mn(VII)$  to  $Mn(IV)$ . Column chromatography of the reaction mixture on this sorbent afforded separation of the target heteroleptic double-decker com-

plex from  $La[(15C5)_4Pc]_2$  and  $La(Pc)_2$ . This approach was later used to prepare the cerium heteroleptic complex  $[(15C5)_4Pc]Ce(Pc)$  [161].

Comparative studies of UV-Vis spectra of the obtained complexes with the corresponding homoleptic counterparts,  $Ln(Pc)_2$  and  $Ln[(15C5)_4Pc]_2$ , suggested that the increased interligand distance in lanthanum complex results in preferential localization of an unpaired electron on the crown-substituted ligand, while the unsubstituted ligand is instead in the dianionic form,  $[(15C5)_4Pc^{\bullet-}]La(Pc^{2-})$ . This asymmetry of electronic density results in higher stability of the anionic form of the heteroleptic double-decker complexes in comparison with the homoleptic compounds.



**Scheme 14.** Synthesis of heteroleptic double-decker complex  $[(15C5)_4Pc]La(Pc)$  [120].

The successful synthesis of  $[(15C5)_4Pc]La(Pc)$  inspired further investigation of the applicability of this method to prepare heteroleptic double-decker complexes of other lanthanides. The idea was to generate an intermediate heteroleptic heteronuclear triple-decker complex  $[(15C5)_4Pc]M(Pc)La(Pc)$  via the reaction of  $H_2[(15C5)_4Pc]$ ,  $La(Pc)_2$  and  $M(acac)_3$ . Low stability of the  $La(III)$  triple-decker complexes could provide selective elimination of a  $La(Pc)^+$  fragment, yielding the target  $[(15C5)_4Pc]M(Pc)$ .

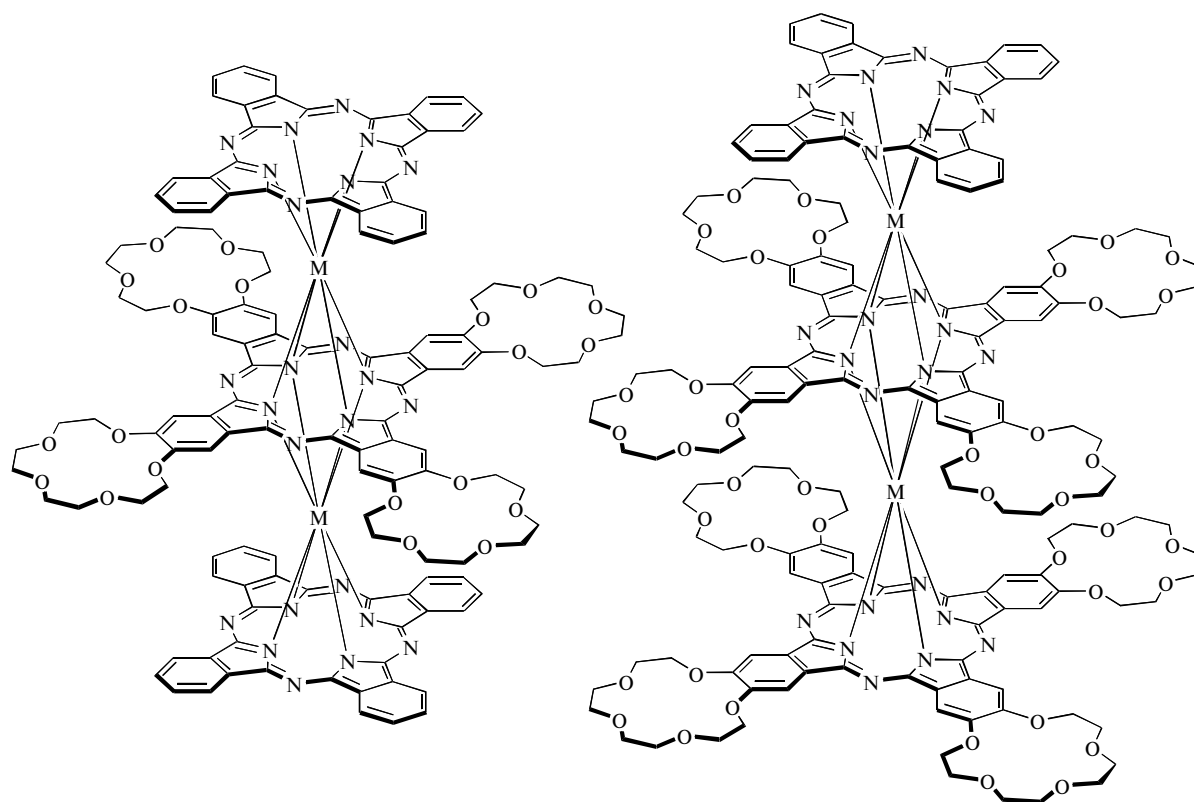
Development of this reaction was initially performed for the compounds with  $M = Sm, Tb, Dy, Tm$  and  $Y$  [162] and later for derivatives with  $Nd, Eu, Ho, Er$  and  $Yb$  [163] (Fig. 7). The target heteroleptic double-decker complexes were obtained only in trace amounts. Unexpectedly, the major products were the heteroleptic homonuclear triple-decker complexes,  $(Pc)M[(15C5)_4Pc]M(Pc)$  and  $[(15C5)_4Pc]M[(15C5)_4Pc]M(Pc)$ , which contained no lanthanum ions. The course of the reaction was monitored using UV-vis spectroscopy. The bands of the initial  $La(Pc)_2$  disappeared immediately after the reaction mixture began to reflux and the band of the intermediate monophthalocyaninate was observed. Recombination of the intermediate monophthalocyaninates then yielded a mixture of triple-decker complexes.

It was supposed that the decomposition of  $La(Pc)_2$  with the formation of  $Pc^{2-}$  species may occur under

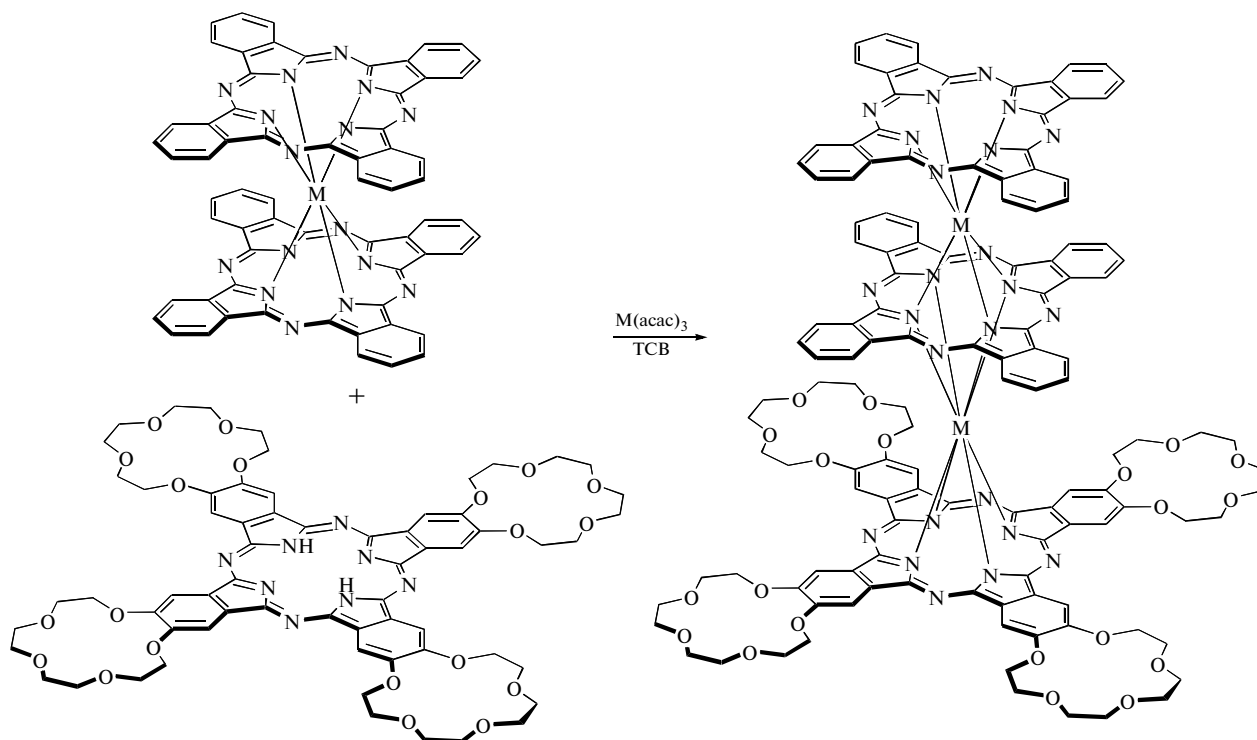
the synthetic conditions [162]. Free lanthanum ion remains in solution and does not participate in subsequent reactions, which was demonstrated by MALDI TOF mass-spectrometry. The reaction time decreased from one hour in the case of  $M = Tm, Yb$  and  $Y$ , to 30 minutes for  $Tb$  and  $Dy$  and to 5–10 minutes for  $Sm$  and  $Nd$ . The  $(Pc)M[(15C5)_4Pc]M(Pc)$  complexes form with a lower yield for the derivatives with the late lanthanides (10–12%) in comparison to those with the middle lanthanides (19–25%). The yield of  $[(15C5)_4Pc]M[(15C5)_4Pc]M(Pc)$  does not significantly change with decrease in the metal ionic radii (39–46%).

In contrast to moisture- and air-sensitive  $Li_2(Pc)$  which is typically used as the  $(Pc^{2-})$  source for synthesis of heteroleptic complexes, [164] the reagent  $La(Pc)_2$  is stable and affords a significant decrease in the reaction time, from several hours, to tens of minutes.

To investigate the influence of the starting  $M(Pc)_2$  stability on the pathway of its reaction with  $H_2[(15C5)_4Pc]$  and  $M(acac)_3 \cdot nH_2O$ , reactions were performed using more stable  $M(Pc)_2$  derivatives,  $M = Sm, Dy, Tm$  and  $Y$ . It resulted in formation of heteroleptic triple-decker complexes with a terminal CRPc ligand— $[(15C5)_4Pc]M(Pc)M(Pc)$  (Scheme 15) [165, 166]. The yields of these complexes decreased upon



**Fig. 7.** Heteroleptic triple-decker complexes  $(\text{Pc})\text{M}[(15\text{C}5)_4\text{Pc}]\text{M}(\text{Pc})$  and  $[(15\text{C}5)_4\text{Pc}]\text{M}[(11\text{C}5)_4\text{Pc}]$ , obtained via the reaction of  $\text{La}(\text{Pc})_2$ ,  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  and  $\text{M}(\text{acac})_3$  in 1-CN,  $\text{M} = \text{Nd}, \text{Sm}, \text{Eu}, \text{Tb}, \text{Dy}, \text{Ho}, \text{Er}, \text{Tm}, \text{Yb}, \text{Y}$  [162, 163].



**Scheme 15.** Synthesis of heteroleptic triple-decker complexes with terminal CRPc ligand [165, 166].

the increase of the metal ionic radii and, therefore, with the decrease of the stability of starting  $M(\text{Pc})_2$ . Due to the presence of strongly paramagnetic lanthanide ions in synthesized triple-decker complexes, their NMR spectra were investigated thoroughly to extract information about structure of these complexes in solution [166, 167].

The addition of monophthalocyaninates to previously synthesized double-decker complexes is a common way to prepare heteroleptic triple-decker complexes with equivalent neighboring ligands [164]. For this purpose, monophthalocyaninates may be also synthesized beforehand or generated in situ in the presence of the double-decker complex. Both approaches were used to synthesize triple-decker complexes bearing CRPc ligands. Among them are the amphiphilic complexes  $[(\text{crown})_4\text{Pc}]\text{Eu}[(\text{crown})_4\text{Pc}]\text{Eu}[(\text{RO})_8\text{Pc}]$  (Fig. 8), which are components of organic-field effect transistors (OFET) [29, 168–170]. These compounds were prepared in moderate yields by the addition of  $[(\text{RO})_8\text{Pc}]\text{Eu}(\text{acac})$ , generated in situ from  $\text{H}_2[(\text{RO})_8\text{Pc}]$  and  $\text{Eu}(\text{acac})_3$  in the presence of  $\text{Eu}[(\text{crown})_4\text{Pc}]_2$ . The influence of the structural factors (size of the crown-ether and length of the alkyl chain) on morphology of the formed thin films was studied to find correlations with characteristics of the OFET devices. For example, two OFETs were made from  $[(\text{crown})_4\text{Pc}]_2\text{Eu}_2[(\text{C}_8\text{H}_{17}\text{O})_8\text{Pc}]$ , where the crown was 12-crown-4 or 18-crown-6. The difference in the size of the crown-ether resulted in different orientations of molecules in films fabricated by the LB technique; in the case of 12C4-triple-decker the formation of an “edge-on” J-aggregate occurs, while 18C6-triple-decker forms cofacial head-to-tail H-aggregates, as evidenced by the observed shifting of the Q-band to the red or to the blue, respectively. As a result, OFETs fabricated from 12C4-triple-decker allow the hole transfer in a direction parallel to the aromatic phthalocyanine rings. In contrast, the devices based on 18C6-triple-decker transfer holes in a direction along the long axis of the assembly composed of face-to-face aggregated molecules.

The addition of monophthalocyaninates to double-decker complexes can be used to introduce *two different* REE ions into one triple-decker molecules, thus yielding heteronuclear compounds [171–174]. The interest in such compounds arises from the fact that being composed of paramagnetic lanthanide ions, they possess unique magnetic properties due to magnetic-dipolar interaction between the lanthanide ions [124, 125, 175].

The addition of  $[(15\text{C}5)_4\text{Pc}]\text{Y}(\text{acac})$  (both generated in situ and synthesized beforehand) gave the first representatives of homoleptic heteronuclear triple-decker tetra-15-crown-5-phthalocyaninato complexes, namely  $\text{LuY}[(15\text{C}5)_4\text{Pc}]_3$  and  $\text{TbY}[(15\text{C}5)_4\text{Pc}]_3$  [119]. These complexes were prepared in 20–30% yield by the reaction between

$\text{Y}[(15\text{C}5)_4\text{Pc}]_2$ ,  $\text{Ln}(\text{acac})_3 \cdot n\text{H}_2\text{O}$  and  $\text{H}_2[(15\text{C}5)_4\text{Pc}]$  in refluxing trichlorobenzene. However, a side reaction between ligand and metal salt together with scrambling processes resulted in formation of both possible homonuclear species. Therefore, the problem of separating the isostructural complexes arose. An improvement in the yield (to 42%) was achieved when the target triple-decker complexes were prepared via addition of previously synthesized lutetium monophthalocyaninate  $[(15\text{C}5)_4\text{Pc}]\text{Lu}(\text{acac})$  to double-decker yttrium complex  $\text{Y}[(15\text{C}5)_4\text{Pc}]_2$ .

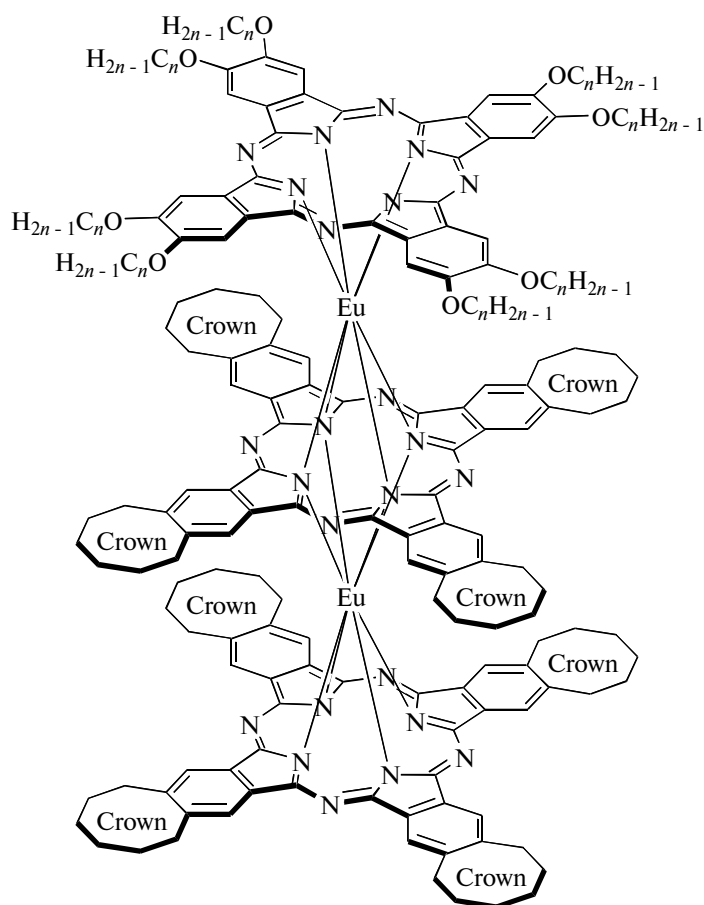
The application of  $\text{La}(\text{Pc})_2$  in the synthesis of heteroleptic complexes was used to develop a new approach to heteronuclear compounds. Generation of  $(\text{Pc})\text{Y}(\text{acac})$  from  $\text{La}(\text{Pc})_2$  and  $\text{Y}(\text{acac})_3$  was performed in the presence of  $\text{Yb}[(15\text{C}5)_4\text{Pc}]_2$ , yielding 61%  $[(15\text{C}5)_4\text{Pc}]\text{Yb}[(15\text{C}5)_4\text{Pc}]\text{Y}(\text{Pc})$ . Using the opposite combination of acetylacetonate and crown-substituted double-decker complex, the isomeric complex  $[(15\text{C}5)_4\text{Pc}]\text{Y}[(15\text{C}5)_4\text{Pc}]\text{Yb}(\text{Pc})$  was obtained in 75% yield (Scheme 16) [176]. The course of the reaction was monitored by UV-Vis spectroscopy. Rapid consumption of double-decker complexes was shown by a decrease in the characteristic absorption around 480 nm. Simultaneously, the fast growth of the new band around 640 nm occurred. Typically, the reactions were complete in ten minutes, almost independent of the introduced lanthanide salt and the double-decker complex. Such a high reaction rate is exceptional in the chemistry of REE tetrapyrrolic complexes, where several hours (from 3 to 18) are required for the formation of sandwich compounds. This indicates the very high reactivity of  $\text{PcM}(\text{acac})$  species generated from  $\text{La}(\text{Pc})_2$ .

### 3. HETEROLEPTIC SANDWICH PORPHYRINATO-CROWN-PHTHALOCYANINATES

Lanthanide double-decker and triple-decker heteroleptic complexes containing porphyrin and phthalocyanine ligands are promising compounds for the development of multibit information storage devices [123], due to the useful electrochemical properties of these compounds. For example, electrochemical studies of the  $[(15\text{C}5)_4\text{Pc}]_2\text{Eu}_2(\text{Ph}_4\text{Por})$  complex, deposited on a gold electrode with simultaneous STM control gave evidence of unique structural rearrangements of the adlayers due to a reversible change in molecular orientation, induced by variation of the potential [177].

Well resolved set of stable reversible redox-states of heteroleptic Por-Pc complexes both in solution and in the solid state can be accessed at relatively low potentials [178]. The introduction of crown-ether substituents into such molecules was expected to improve their electrochemical properties, as well as facilitate LB film formation and finally, to afford tuning of physical-





**Fig. 8.** Heteroleptic triple-decker complexes  $[(\text{crown})_4\text{Pc}]\text{Eu}[(\text{crown})_4\text{Pc}]\text{Eu}[(\text{RO})_8\text{Pc}]$  prepared for application in OFET devices. Crown = 12-crown-4, 18-crown-6,  $n = 8$  [29]; crown = 15-crown-5,  $n = 4, 6, 8, 10, 12$  [169, 170].

chemical characteristics of Por-CRPc composites by means of molecular recognition.

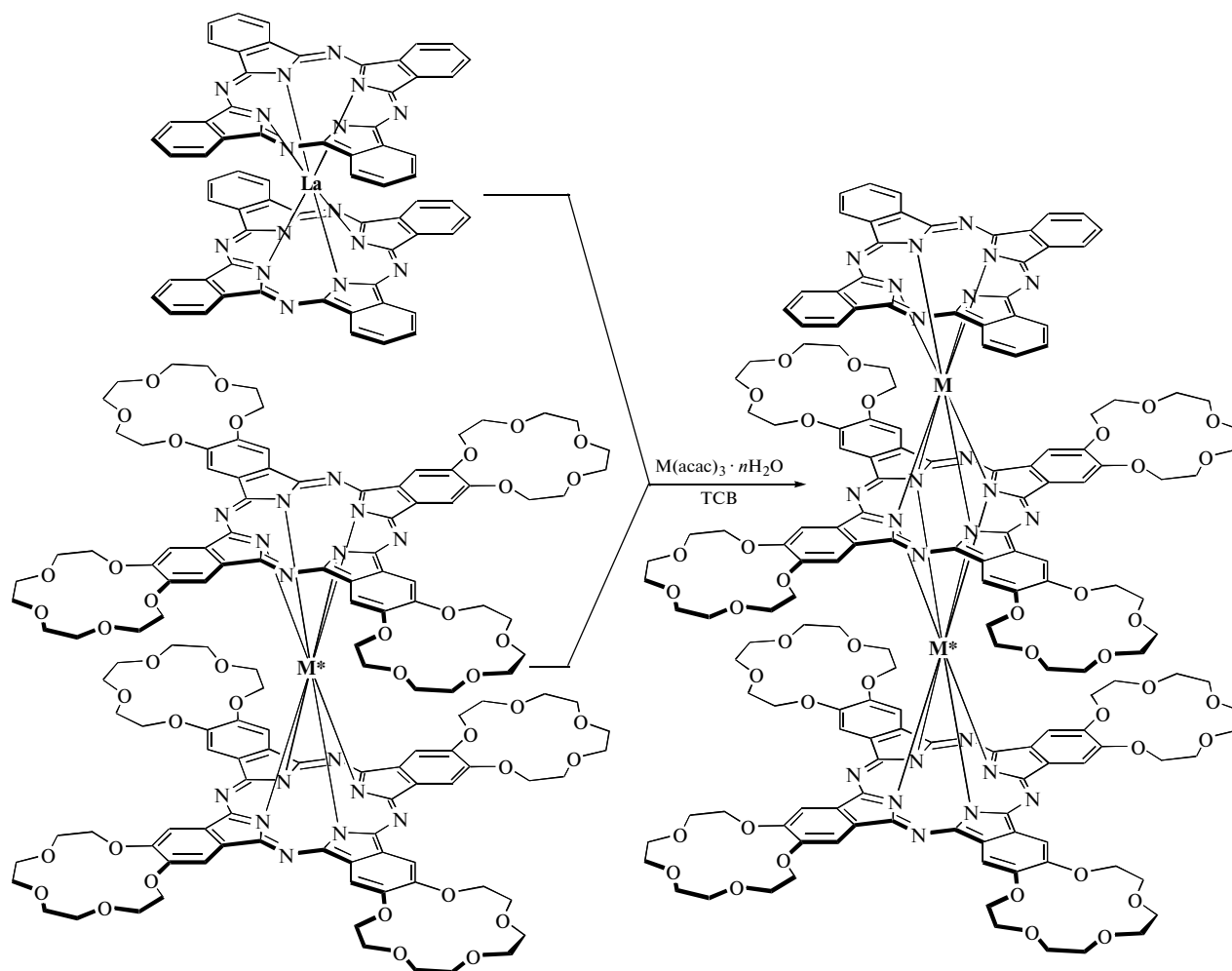
The first examples of Por-CRPc heteroleptic complexes were prepared via the “raise-by-one-storey” approach starting from  $\text{H}_2(\text{Ph}_4\text{Por})$  [179]. Its reaction with  $\text{Eu}(\text{acac})_3 \cdot n\text{H}_2\text{O}$  in 1-octanol generated monoporphyrate  $(\text{Ph}_4\text{Pc})\text{Eu}(\text{acac})$ , which acted as a template agent for preparation of the heteroleptic complex  $[(15\text{C}5)_4\text{Pc}]\text{Eu}(\text{Ph}_4\text{Por})$ . Addition of one more porphyrin deck to this complex was achieved by its reaction with  $(\text{Ph}_4\text{Por})\text{Eu}(\text{acac})$ , generated in TCB. The addition of  $(\text{Ph}_4\text{Por})\text{Eu}(\text{acac})$  to  $\text{Eu}[(15\text{C}5)_4\text{Pc}]_2$  yielded  $[(15\text{C}5)_4\text{Pc}]\text{Eu}[(15\text{C}5)_4\text{Pc}]\text{Eu}(\text{Ph}_4\text{Por})$  (Scheme 17).

The abovementioned pathway to heteroleptic triple-decker complexes with neighboring CRPc ligands via addition of a monoporphyrate to a crown-substituted double-decker complex was used to prepare amphiphilic complexes as components of OFET devices. For this purpose, *meso*-tetradecylporphyrin,  $\text{H}_2[(\text{C}_{12}\text{H}_{25})_4\text{Por}]$ , or *meso*-tetra(4-pentylphenyl)porphyrin,  $\text{H}_2[(4\text{-PentOPh})_4\text{Por}]$ , were treated with excess  $\text{Eu}(\text{acac})_3$  to obtain monoporphyrates, which further reacted with  $\text{Eu}[(15\text{C}5)_4\text{Pc}]_2$  to yield 38 and

42% of the target heteroleptic complexes, respectively [180].

Generally, heteroleptic Por-Pc complexes are synthesized by stepwise procedures, which require the preparation of monoporphyrates. That is why these methods are typically applied only to middle and late lanthanides, which form stable-enough intermediates. In the case of La and Pr such reactions are characterized by low yields. Therefore, the “one-pot” approaches without isolation of unstable intermediates were expected to solve this problem [181–183]. With this aim, the “one-pot” reaction of  $\text{H}_2(\text{An}_4\text{Por})$  ( $\text{An} = 4\text{-MeOPh}$ ) with DCB15C5,  $\text{Ln}(\text{acac})_3$ , and DBU in 1-octanol was studied. The studied lanthanides were La, Ce, and Pr [181], followed by Nd, Sm, Eu [182] and Tb [113]. The principal difference between this procedure and those previously described is that it does not require separation and purification of monoporphyric species. In contrast, the monoporphyrate is generated in situ as a result of the equilibration interaction.

Monitoring of the reaction pathway by MALDI TOF MS allowed the presumption of a stepwise mechanism for this process. After one hour of reflux, the



**Scheme 16.** Synthesis of heteronuclear triple-decker complexes,  $M \neq M = \text{Y, Yb}$  [176].

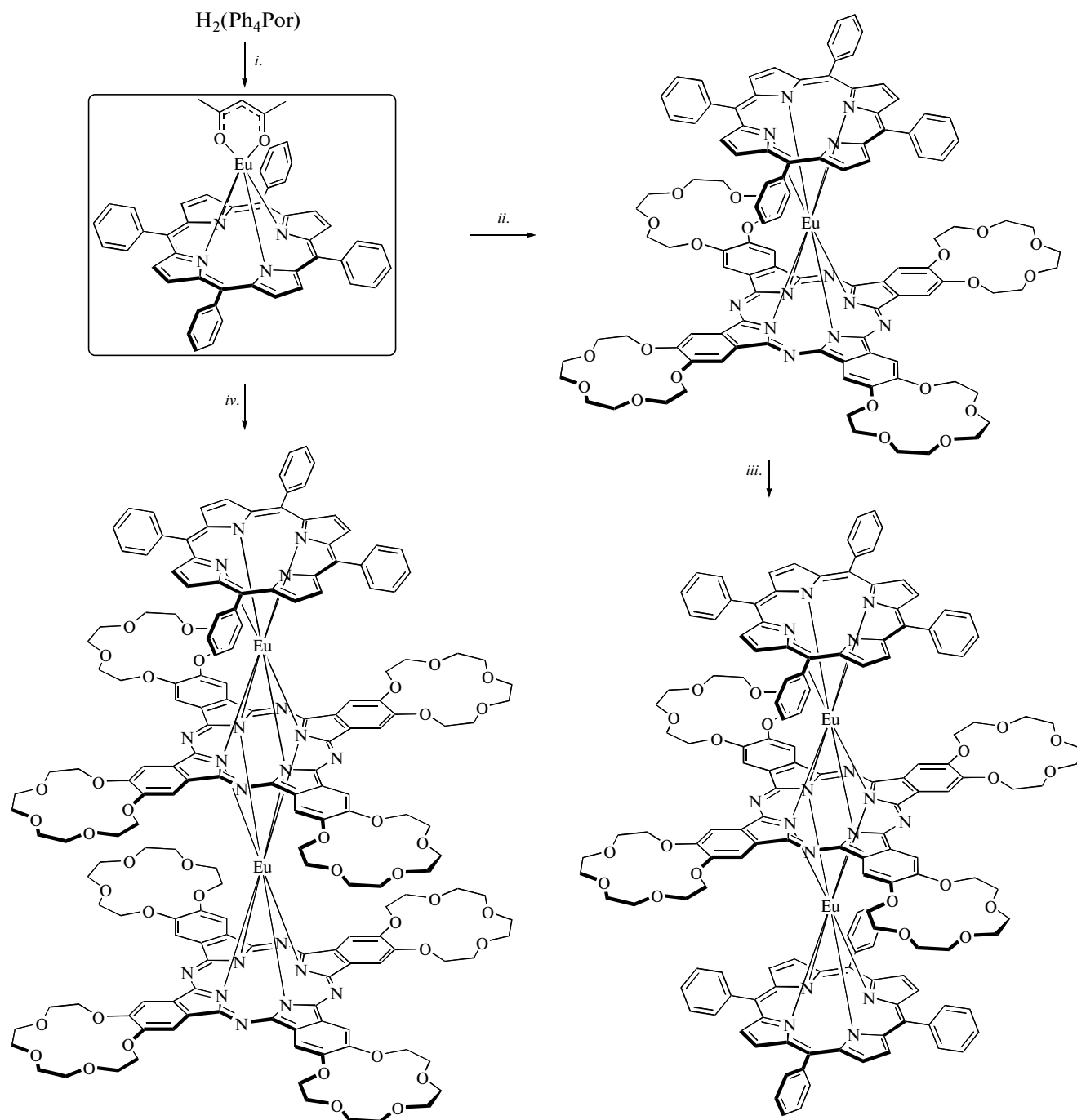
lanthanide monoporphyrate ( $\text{An}_4\text{Por}$ ) $\text{Ln}(\text{acac})$  and the heteroleptic double-decker complex ( $\text{An}_4\text{Por}$ ) $\text{Ln}[(15\text{C}5)_4\text{Pc}]$  were observed in the reaction mixture. After two hours, traces of the triple-decker complex ( $\text{An}_4\text{Por}$ ) $\text{Ln}[(15\text{C}5)_4\text{Pc}]\text{Ln}(\text{An}_4\text{Por})$  were detected. During further reflux, the ratio between the double-decker and the triple-decker complex gradually changed with the rise of triple-decker amount. The process was terminated after 18 hours of reflux when the monoporphyrate was no longer detected and the double-decker/triple-decker ratio had stabilized. Notably, neither  $\text{H}_2[(15\text{C}5)_4\text{Pc}]_2$  nor its homoleptic double-decker complexes were observed.

The regioselective formation of one triple-decker complex as a single symmetrical isomer in this reaction suggests that the attack of the monoporphyrate occurs regioselectively on the Pc side of the intermediate double-decker molecule. To explain this regioselectivity, quantum-chemical calculations were performed for the model complex ( $\text{An}_4\text{Por}$ ) $\text{Y}[(\text{MeO})_8\text{Pc}]$ , which indicated asymmetric distribution of electronic density

over tetrapyrrole ligands in the double-decker complex—66.2% of the HOMO was localized on the Pc and 33.8% on the Por ligand [181]. Solution structure of synthesized complexes was studied by NMR spectroscopy due to the presence of paramagnetic lanthanide ions [184].

The influence of various factors on the “one-pot” synthesis of the heteroleptic Por-CRPc complexes of La, Pr, Nd and Eu was studied using 4-bromophenyl-substituted porphyrin,  $\text{H}_2(\text{BrPh}_4\text{Por})$ . Among the examined factors were the ratio of reagents and the temperature of the synthesis. As a result, optimal conditions were found to obtain triple-decker complexes as single reaction products [183].

Previously it was demonstrated that, in the case of cerium, only a double-decker complex could be obtained with CRPc ligands, and the formation of triple-decker complex was never observed because of the stabilization of the quadrivalent state of the Ce ion. However, when the mixed-ligand complexes containing both Por and Pc ligands were investigated,



**Scheme 17.** Synthesis of heteroleptic porphyrinato-crown-phthalocyaninato complexes: *i.*  $\text{Eu}(\text{acac})_3$ ,  $n\text{H}_2\text{O}$ , 1-octanol or TCB,  $200^\circ\text{C}$ , 4 hours (without isolation); *ii.* DCB15C5, DBU, 1-octanol,  $140^\circ\text{C}$ , 12 hours (yield 43%); *iii.*  $[(15\text{C}5)_4\text{Pc}]\text{Eu}(\text{Ph}_4\text{Por})$ , TCB,  $210^\circ\text{C}$ , 4 hours (yield 50%); *iv.*  $\text{Eu}[(15\text{C}5)_4\text{Pc}]_2$ , TCB,  $210^\circ\text{C}$ , 4 hours (yield 42%) [179].

both the double-decker and triple-decker complexes were isolated,  $(\text{An}_4\text{Por})\text{Ce}[(15\text{C}5)_4\text{Pc}]$  and  $(\text{An}_4\text{Por})\text{Ce}[(15\text{C}5)_4\text{Pc}]\text{Ce}(\text{An}_4\text{Por})$ . The oxidation state of the Ce ion in these complexes was investigated by UV-Vis studies in comparison with other available isostructural complexes [182].

## SUMMARY

Starting from attempts to improve the solubility of phthalocyanines and to control their aggregation, two last decades of crown-phthalocyanine chemistry have been marked by comprehensive studies of synthesis of novel CRPc's, their assembly induced by various fac-

tors, fabrication of thin films, incorporation into optoelectronic devices, and synthesis of biomimetic models. The accumulated knowledge allows one to develop the link between laboratory chemical synthesis and development of innovative materials. It was demonstrated that the particular feature of CRPc's is the fortunate combination of advantages of two classes of macrocyclic compounds. Therefore, the crownphthalocyanine's containing materials exhibit synergistic properties, giving rise to multifunctional devices.

The obvious potential of these compounds in sensing applications, photovoltaic and photorefracting devices, opto- and iono-electronics suggests novel advances in this rich area in the near future.

### ACKNOWLEDGMENTS

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### REFERENCES

1. A. W. Snow, in *The Porphyrin Handbook*, Ed. by K. M. Kadish, K. M. Smith, and R. Guilard (Academic Press, New York, 2003), Vol. 17, p. 129.
2. E. Iengo, E. Zangrando, and E. Alessio, *Acc. Chem. Res.* **39**, 841 (2006).
3. V. V. Borovkov, G. A. Hembury, and Y. Inoue, *Acc. Chem. Res.*, 449 (2004).
4. C. M. Drain and A. Varotto, I. Radivojevic, *Chem. Rev.* **109**, 1630 (2009).
5. *Non-Covalent Multi-Porphyrin Assemblies*, Ed. by E. Alessio (Springer, Berlin/Heidelberg, 2006), vol. 121.
6. F. D'Souza and O. Ito, *Coord. Chem. Rev.* **249**, 1410 (2005).
7. I. P. Beletskaya, V. S. Tyurin, A. Y. Tsivadze, et al., *Chem. Rev.* **109**, 1659 (2009).
8. A. Y. Tsivadze, *Russ. Chem. Rev.* **73**, 5 (2004).
9. P. Even and B. Boitrel, *Coord. Chem. Rev.* **250**, 519 (2006).
10. Y. G. Gorbunova, A. G. Martynov, and A. Y. Tsivadze, in *Handbook of Porphyrin Science*, Ed. by K. M. Kadish, K. M. Smith, and R. Guilard (World Scientific, 2012), vol. 24, p. 271.
11. A. G. Martynov, Y. G. Gorbunova, S. E. Nefedov, et al., *Eur. J. Org. Chem.* **2012**, 6888 (2012).
12. C. F. Van Nostrum, S. J. S. J. Picken, and R. J. M. Nolte, *Angew. Chem. Int. Ed.* **33**, 2173 (1994).
13. C. F. van Nostrum, S. J. Picken, A.-J. Schouten, et al., *J. Am. Chem. Soc.* **117**, 9957 (1995).
14. H. Engelkamp, S. Middelbeek, and R. J. M. Nolte, *Science* **284**, 785 (1999).
15. H.-W. Liu, C.-F. Chen, M. Ai, et al., *Tetrahedron: Asymmetry* **11**, 4915 (2000).
16. H. Liu, Y. Liu, M. Liu, et al., *Tetrahedron Lett.* **42**, 7083 (2001).
17. M. V. Martínez-Díaz, N. S. Fender, M. S. Rodríguez-Morgade, et al., *J. Mater. Chem.* **12**, 2095 (2002).
18. J. Sly, P. Kasák, E. Gomar-Nadal, et al., *Chem. Commun.*, No. 10, 1255 (2005).
19. H. Kantekin, M. Rakap, Y. Gök, et al., *Dye. Pigm.* **74** (1), 21 (2007).
20. A. R. Koray, V. Ahsen, and Ö. Bekâroğlu, *J. Chem. Soc., Chem. Commun.*, No. 12, 932 (1986).
21. N. Kobayashi and Y. Nishiyama, *J. Chem. Soc., Chem. Commun.*, No. 19, 1462 (1986).
22. R. Hendriks, O. E. Sielcken, W. Drenth, et al., *J. Chem. Soc., Chem. Commun.*, No. 19, 1464 (1986).
23. N. Kobayashi and A. B. P. Lever, *J. Am. Chem. Soc.* **109**, 7433 (1987).
24. O. E. Sielcken, M. M. Van Tilborg, M. F. M. Roks, et al., *J. Am. Chem. Soc.* **109**, 4261 (1987).
25. V. Ahsen, E. Yilmazer, M. Ertas, et al., *J. Chem. Soc., Dalt. Trans.*, No. 2, 401 (1988).
26. R. F. Ziolo and M. Extine, *Inorg. Chem.* **20**, 2709 (1981).
27. R. F. Ziolo, W. H. H. Guenther, and J. M. Troup, *J. Am. Chem. Soc.* **103**, 4629 (1981).
28. Y. G. Gorbunova, O. Y. Komarova, S. V. Demin, et al., *Russ. J. Coord. Chem.* **23**, 516 (1997).
29. Y. Gao, P. Ma, Y. Chen, et al., *Inorg. Chem.* **48**, 45 (2008).
30. V. Ahsen, E. Yilmazer, Ö. Bekâroğlu, et al., *Macromol. Rapid Commun.* **8**, 243 (1987).
31. V. Ahsen, E. Yilmazer and Ö. Bekâroğlu, *Macromol. Chem.* **189**, 2533 (1988).
32. D. M. Guldi, J. Ramey, M. V. Martínez-Díaz, et al., *Chem. Commun.*, No. 23, 2774 (2002).
33. A. G. Martynov, K. P. Birin, Y. G. Gorbunova, et al., *Macroheterocycles* **6**, 23 (2013).
34. T. V. Trofimova, Y. G. Gorbunova, E. O. Tolkacheva, et al., *Russ. J. Coord. Chem.* **25**, 402 (1999).
35. N. Kobayashi and M. Opallo, *Heterocycles* **30**, 389 (1990).
36. E. Musluoglu, A. Gürek, V. Ahsen, et al., *Chem. Ber.* **125**, 2337 (1992).
37. N. Kobayashi, P. Janda, and A. B. P. Lever, *Inorg. Chem.* **31**, 5172 (1992).
38. V. Ahsen, E. Yilmazer, A. Gürek, et al., *Helv. Chim. Acta* **71**, 1616 (1988).
39. O. E. Sielcken, L. A. Van de Kuil, W. Drenth, et al., *J. Chem. Soc., Chem. Commun.*, No. 18, 1232 (1988).
40. O. E. Sielcken, L. A. Van de Kuil, W. Drenth, et al., *J. Am. Chem. Soc.* **112**, 3086 (1990).
41. S. Lee and H. Suh, *Bull. Korean Chem. Soc.* **20**, 991 (1999).
42. N. Kobayashi, T. Ishizaki, K. Ishii, et al., *J. Am. Chem. Soc.* **121**, 9096 (1999).
43. M. S. Rodríguez-Morgade, G. de la Torre, and T. Torres, in *The Porphyrin Handbook*, Ed. by K. M. Kadish, K. M. Smith, and R. Guilard (Academic Press, New York, 2003), Vol. 15, p. 125.
44. K. P. Birin, V. N. Chugunov, A. A. Krapivenko, et al., *Macroheterocycles* **7**, 153 (2014).
45. T. Fukuda, I. Sugita, N. Kobayashi, *Chem. Commun.*, No. 23, 3449 (2009).

46. J. Vacus, G. Memetzidis, P. Doppelt, et al., *J. Chem. Soc., Chem. Commun.* **302**, 697 (1994).
47. D. Pernin and J. Simon, *Mol. Cryst. Liq. Cryst.* **355**, 457 (2001).
48. D. Pernin, K. Haberoth, and J. Simon, *J. Chem. Soc., Perkin Trans.* **1**, 1265 (1997).
49. A. G. Martynov, M. V Panova, Y. G. Gorbunova, et al., *Macrocyclic Chem.* **7**, 47 (2014).
50. S. O. Andryushkevich, K. P. Birin, Y. G. Gorbunova, et al., *Prot. Met.* **47**, 494 (2011).
51. E. O. Tolkacheva, S. V. Selivanova, O.Y. Komarova, et al., *Russ. J. Inorg. Chem.* **43**, 209 (1998).
52. D. Reddy and T. K. Chandrashekar, *Polyhedron* **12**, 627 (1993).
53. E. O. Tolkacheva, L. I. Demina, A. Y. Tsivadze, et al., *Russ. J. Inorg. Chem.* **40**, 432 (1995).
54. L. A. Lapkina, N. Y. Konstantinov, V. E. Larchenko, et al., *J. Porphyrins Phthalocyanines* **13**, 859 (2009).
55. A. V. Vannikov, A. D. Grishina, Y. G. Gorbunova, et al., *Polym. Sci. A* **53**, 1069 (2011).
56. L. A. Lapkina, Y. G. Gorbunova, D.O. Gil, et al., *J. Porphyrins Phthalocyanines* **17**, 564 (2013).
57. E. G. Vakoulovskaya, V. V. Shental, L. V. Oumnova, et al., *Proc. SPIE* **4952**, 149 (2003).
58. A. Y. Tsivadze, E. O. Tolkacheva, L. I. Demina, et al., *Patent RU 2079499* (1997).
59. T. Rawling and A. McDonagh, *Coord. Chem. Rev.* **251**, 1128 (2007).
60. H. Bertagnolli, A. Weber, W. Hörner, et al., *Inorg. Chem.* **36**, 6397 (1997).
61. P. Stuzhin, S. I. Vagin, and M. Hanack, *Inorg. Chem.* **37**, 2655 (1998).
62. M. Weidemann, H. Hückstädt, and H. Homborg, *Z. Anorg. Alg. Chem.* **624**, 846 (1998).
63. T. Rawling, H. Xiao, S.-T. Lee, et al., *Inorg. Chem.* **46**, 2805 (2007).
64. M. T. Brumbach, A. K. Boal, and D. R. Wheeler, *Langmuir* **25**, 10685 (2009).
65. M. S. Rodríguez-Morgade, M. E. Plonska-Brzezinska, A. J. Athans, et al., *J. Am. Chem. Soc.* **131**, 10484 (2009).
66. J. P. Collman, C. E. Barnes, P. N. Swepston, et al., *J. Am. Chem. Soc.* **106**, 3500 (1984).
67. A. Capobianchi, A. M. Paoletti, G. Pennesi, et al., *Inorg. Chem.* **33**, 4635 (1994).
68. R. Caminiti, M. P. Donzello, C. Ercolani, et al., *Inorg. Chem.* **38**, 3027 (1999).
69. A. Kienast, C. Bruhn, and H. Homborg, *Z. Anorg. Alg. Chem.* **623**, 967 (1997).
70. G. Rossi, M. Gardini, G. Pennesi, et al., *J. Chem. Soc., Dalt. Trans.*, No. 1, 193 (1989).
71. A. Capobianchi, G. Pennesi, A. M. Paoletti, et al., *Inorg. Chem.* **35**, 4643 (1996).
72. M. Hanack, S. Kamenzin, C. Kamenzin, et al., *Synth. Met.* **110**, 93 (2000).
73. W. Zhao, B. Tong, Y. Pan, et al., *Langmuir* **25**, 11796 (2009).
74. G. de la Torre, C. G. Claessens, and T. Torres, *Chem. Commun.*, No. 20, 2000 (2007).
75. A. Morandeira, I. López-Duarte, B. O'Regan, et al., *J. Mater. Chem.* **19**, 5016 (2009).
76. M. V. Martínez-Díaz, G. de la Torre, and T. Torres, *Chem. Commun.* **46**, 7090 (2010).
77. Y. Y. Enakieva, Y. G. Gorbunova, S. G. Sakharov, et al., *Russ. J. Inorg. Chem.* **47**, 1815 (2002).
78. S. Omiya, M. Tsutsui, E. F. Meyer, et al., *Inorg. Chem.* **19**, 134 (1980).
79. Y. G. Gorbunova, Y. Y. Enakieva, S. G. Sakharov, et al., *J. Porphyrins Phthalocyanines* **7**, 795 (2003).
80. Y. Y. Enakieva, Y. G. Gorbunova, S. E. Nefedov, et al., *Mendeleev Commun.* **14** (5), 193 (2004).
81. Y. G. Gorbunova, Y. Y. Enakieva, S. G. Sakharov, *Russ. Chem. Bull., Int. Ed.* **53**, 74 (2004).
82. A. D. Grishina, Y. G. Gorbunova, L. Y. Pereshivko, et al., *Prot. Met.* **45**, 535 (2009).
83. L. A. Khanova, L. I. Krishtalik, Y. Y. Enakieva, et al., *Russ. J. Electrochem.* **43**, 1350 (2007).
84. R. Polley, M. Hanack, *Synthesis* **1997**, 295 (1997).
85. L. Y. Pereshivko, A. D. Grishina, Y. G. Gorbunova, et al., *High En. Chem.* **43**, 543 (2010).
86. A. D. Grishina, Y. G. Gorbunova, V. I. Zolotarevsky, et al., *J. Porphyrins Phthalocyanines* **13**, 92 (2009).
87. A. D. Grishina, V. I. Zolotarevsky, Y. G. Gorbunova, et al., *Russ. J. Phys. Chem. A* **83**, 1907 (2009).
88. A. D. Grishina, Y. G. Gorbunova, Y. Y. Enakieva, et al., *High En. Chem.* **42**, 297 (2008).
89. Y. Y. Enakieva, Y. G. Gorbunova, A. Y. Tsivadze, et al., *J. Porphyrins Phthalocyanines* **11**, 883 (2007).
90. J. W. Buchler, D. K. P. Ng, in *The Porphyrin Handbook*, Ed. by K. M. Kadish, K. M. Smith, and R. Guilard (Academic Press, New York, 2000), Vol. 3, p. 245.
91. R. Weiss, J. Fischer, in *The Porphyrin Handbook*, Ed. by K. M. Kadish, K. M. Smith, and R. Guilard (Academic Press, New York, 2003), Vol. 16, p. 171.
92. G. A. Corker, B. Grant, and N. J. Clecak, *J. Electrochem. Soc.* **126**, 1339 (1979).
93. A. T. Chang and J.-C. Marchon, *Inorg. Chim. Acta* **53**, L241 (1981).
94. M. L'Her, Y. Cozien, and J. Courtot-Coupez, *J. Electroanal. Chem.* **157**, 183 (1983).
95. P. Turek, P. Petit, J. J. Andre, et al., *J. Am. Chem. Soc.* **109**, 5119 (1987).
96. P. N. Moskalev, *Sov. J. Coord. Chem. (Engl. Transl.)* **16**, 77 (1990).
97. H. Konami, M. Hatano, N. Kobayashi, et al., *Chem. Phys. Lett.* **165**, 397 (1990).
98. A. Iwase, C. Harnood, and Y. Kameda, *J. Alloy. Compd.* **192** (1–2), 280 (1993).
99. F. Guyon, A. Pondaven, J.-M. Kerbaol, et al., *Inorg. Chem.* **37**, 569 (1998).
100. P. Bassoul and T. Toupance, *J. Simon, Sens. Actuators, B* **26** (1–3), 150 (1995).
101. M. L. Rodríguez-Méndez, J. Souto, J. de Saja-Gonzalez, et al., *Sens. Actuators, B* **31** (1–2), 51 (1996).
102. R. Rousseau, R. Aroca, and M. L. Rodríguez-Méndez, *J. Mol. Struct.* **356**, 49 (1995).

103. N. Ishikawa and Y. Kaizu, *J. Phys. Chem.* **100**, 8722 (1996).
104. P. Zhu, N. Pan, C. Ma, et al., *Eur. J. Inorg. Chem.*, No. 3, 518 (2004).
105. A. G. Martynov, Y. G. Gorbunova, and A. Y. Tsivadze, *Prot. Met.* **47**, 465 (2011).
106. M. Bardin, E. Bertounesque, V. Plichon, et al., *J. Electroanal. Chem.* **271**, 173 (1989).
107. T. Toupance, V. Ahsen, and J. Simon, *J. Am. Chem. Soc.* **116**, 5352 (1994).
108. I. S. Kirin, P.N. Moskalev, and Y. A. Makashev, *Rus. J. Inorg. Chem.* **12** 369 (1967).
109. K. Kasuga, M. Ando, H. Morimoto, et al., *Chem. Lett.* **2**, 1095 (1986).
110. E. O. Tolkacheva, A. Y. Tsivadze, Z. G. Bitiev, et al., *Russ. J. Inorg. Chem.* **40**, 949 (1995).
111. L. A. Lapkina, E. Niskanen, H. Rönkkömäki, et al., *Russ. J. Coord. Chem.* **24**, 598 (1998).
112. C. I. Troyanov, L. A. Lapkina, V. E. Larchenko, et al., *Dokl. Chem.* **367**, 192 (1999).
113. K. P. Birin, Y. G. Gorbunova, A. Y. Tsivadze, *Dalt. Trans.*, **41**, 23 (2012).
114. L. A. Lapkina, V. E. Larchenko, E. O. Tolkacheva, et al., *Russ. J. Inorg. Chem.* **43**, 901 (1998).
115. L. A. Lapkina, E. Niskanen, H. Rönkkömäki, et al., *J. Porphyrins Phthalocyanines* **4**, 588 (2000).
116. Y. G. Gorbunova, L. A. Lapkina, A. G. Martynov, et al., *Russ. J. Coord. Chem.* **30**, 245 (2004).
117. A. Y. Tsivadze, E. O. Tolkacheva, V. E. Larchenko, et al., *Patent RU 2118326* (1998).
118. I. V. Nefedova, Y. G. Gorbunova, S. G. Sakharov, et al., *Russ. J. Inorg. Chem.* **50**, 165 (2005).
119. I. V. Nefedova, Y. G. Gorbunova, S. G. Sakharov, et al., *Mendeleev Commun.* **16**, 67 (2006).
120. A. G. Martynov and Y. G. Gorbunova, *Inorg. Chim. Acta* **360**, 122 (2007).
121. A. A. Sinelshchikova, Y. G. Gorbunova, L. A. Lapkina, et al., *Russ. J. Inorg. Chem.* **56**, 1370 (2011).
122. I. V. Zhukov, L. A. Lapkina, Y. G. Gorbunova, et al., *J. Porphyrins Phthalocyanines* **9**, 1 (2005).
123. J. S. Lindsey and D. F. Bocian, *Acc. Chem. Res.* **44**, 638 (2011).
124. N. Ishikawa, T. Iino, and Y. Kaizu, *J. Am. Chem. Soc.* **124**, 11440 (2002).
125. N. Ishikawa, in *Functional Phthalocyanine Molecular Materials*, Ed. by J. Jiang (Springer, Berlin, Heidelberg, 2010), Vol. 135, p. 211.
126. K. P. Birin, Y. G. Gorbunova, and A. Y. Tsivadze, *Russ. J. Inorg. Chem.* **52**, 191 (2007).
127. A. G. Martynov, Y. G. Gorbunova, I. V. Khrapova, et al., *Russ. J. Inorg. Chem.* **47**, 1616 (2002).
128. A. G. Martynov, I. V. Nefedova, Y. G. Gorbunova, et al., *Mendeleev Commun.* **17**, 66 (2007).
129. L. A. Lapkina, S. G. Sakharov, N. Y. Konstantinov, et al., *Russ. J. Inorg. Chem.* **52**, 1758 (2007).
130. J. Jiang, W. Liu, K.-W. Poon, et al., *Eur. J. Inorg. Chem.*, No. 1, 205 (2000).
131. W. Liu, J. Jiang, D. Du, et al., *Aust. J. Chem.* **53**, 131 (2000).
132. Y. Bian, D. Wang, R. Wang, et al., *New J. Chem.* **27**, 844 (2003).
133. Y. Bian, J. Jiang, Y. Tao, et al., *J. Am. Chem. Soc.* **125**, 12257 (2003).
134. P. Zhu, F. Lu, N. Pan, et al., *Eur. J. Inorg. Chem.*, No. 3, 510 (2004).
135. K. P. Birin, Y. G. Gorbunova, and A. Y. Tsivadze, *J. Porphyrins Phthalocyanines* **10**, 931 (2006).
136. K. P. Birin, S. E. Nefedov, Y. G. Gorbunova, et al. (unpublished).
137. L. A. Lapkina, Y. G. Gorbunova, S. E. Nefedov, et al., *Russ. Chem. Bull., Int. Ed.* **52**, 1633 (2003).
138. P. Roisin, J. D. Wright, R. J. M. Nolte, et al., *J. Mater. Chem.* **2**, 131 (1992).
139. J. D. Wright, P. Roisin, G. P. Rigby, et al., *Sens. Actuators, B* **13**, 276 (1993).
140. J. D. Wright, P. Roisin, G. Patricia Rigby, et al., *Sens. Actuators, B* **16**, 301 (1993).
141. M. J. Jory, P. S. Cann, J. R. Sambles, *J. Phys. D: Appl. Phys.* **27**, 169 (1994).
142. X. Li, *Thin Solid Films* **324**, 277 (1998).
143. X. Li, Y. Chen, H. Xu, *J. Mater. Chem.* **9**, 1415 (1999).
144. M. L. Rodríguez-Méndez, J. Souto, and J. A. De Saja, *J. Raman Spectrosc.* **26**, 693 (1995).
145. T. Toupance, V. Plichon, and J. Simon, *New J. Chem.* **23**, 1001 (1999).
146. S. L. Selektor, V. V. Arslanov, Y. G. Gorbunova, et al., *J. Porphyrins Phthalocyanines*, **12**, 1154 (2008).
147. S. L. Selektor, A. V. Shokurov, V. V. Arslanov, et al., *J. Phys. Chem. C* **118**, 4250 (2014).
148. D. K. P. Ng and J. Jiang, *Chem. Soc. Rev.* **26**, 433 (1997).
149. V. E. Pushkarev, L. G. Tomilova, Y. V. Tomilov, *Russ. Chem. Rev.* **77**, 875 (2008).
150. J. Jiang and D. K. P. Ng, *Acc. Chem. Res.* **42**, 79 (2009).
151. N. Ishikawa and Y. Kaizu, *Chem. Phys. Lett.* **203**, 472 (1993).
152. N. Ishikawa and Y. Kaizu, *Chem. Phys. Lett.* **228**, 625 (1994).
153. N. Ishikawa, *Chem. Phys. Lett.* **236**, 50 (1995).
154. Y. G. Gorbunova, E. O. Tolkacheva, and A. Y. Tsivadze, *Russ. J. Coord. Chem.* **22**, 884 (1996).
155. F. Steybe and J. Simon, *New J. Chem.* **22**, 1305 (1998).
156. T. Thami, C. Chassenieux, C. Fretigny, et al., *J. Porphyrins Phthalocyanines* **06**, 563 (2002).
157. N. Ishikawa and Y. Kaizu, *Chem. Lett.* **1998**, 183 (1998).
158. N. Sheng, R. Li, C.-F. Choi, et al., *Inorg. Chem.* **45**, 3794 (2006).
159. N. Sheng, Y. Zhang, H. Xu, et al., *Eur. J. Inorg. Chem.*, No. 20, 3268 (2007).
160. X. Cai, N. Sheng, Y. Zhang, et al., *Spectrochim. Acta, Part A* **72**, 627 (2009).
161. S. L. Selektor, A. V. Shokurov, V. V. Arslanov, et al., *Russ. J. Electrochem.* **48**, 218 (2012).
162. A. G. Martynov, O. V. Zubareva, Y. G. Gorbunova, et al., *Eur. J. Inorg. Chem.*, No. 30, 4800 (2007).

163. A. G. Martynov, E. A. Safonova, Y. G. Gorbunova, et al., *Russ. J. Inorg. Chem.* **55**, 347 (2010).
164. W. Liu, J. Jiang, N. Pan, et al., *Inorg. Chim. Acta* **310**, 140 (2000).
165. A. G. Martynov, O. V. Zubareva, Y. G. Gorbunova, et al., *Inorg. Chim. Acta* **362**, 11 (2009).
166. A. G. Martynov and Y. G. Gorbunova, *Polyhedron* **29**, 391 (2010).
167. A. G. Martynov, Y. G. Gorbunova, and A. Y. Tsivadze, *Dalt. Trans.* **40**, 7165 (2011).
168. Y. Zhang, X. Cai, Y. Bian, et al., in *Functional Phthalocyanine Molecular Materials*, Ed. by J. Jiang (Springer, Berlin, Heidelberg, 2010), vol. 135, pp. 275–322.
169. Y. Chen, R. Li, R. Wang, et al., *Langmuir* **23**, 12549 (2007).
170. Y. Chen, M. Bouvet, T. Sizun, et al., *Phys. Chem. Chem. Phys.* **12**, 12851 (2010).
171. P. Zhu, N. Pan, R. Li, et al., *Chem. Eur. J.* **11**, 1425 (2005).
172. N. Ishikawa, T. Iino, and Y. Kaizu, *J. Phys. Chem. A* **106**, 9543 (2002).
173. N. Ishikawa, T. Iino, and Y. Kaizu, *J. Phys. Chem. A* **107**, 7879 (2003).
174. V. E. Pushkarev, E. V. Shulishov, Y. V. Tomilov, et al., *Tetrahedron Lett.* **48**, 5269 (2007).
175. N. Ishikawa, S. Otsuka, and Y. Kaizu, *Angew. Chem. Int. Ed.* **44**, 731 (2005).
176. A. Y. Tsivadze, A. G. Martynov, M. A. Polovkova, et al., *Russ. Chem. Bull., Int. Ed.*, No. 11, 2216 (2011).
177. S. Yoshimoto, T. Sawaguchi, W. Su, et al., *Angew. Chem. Int. Ed.* **46**, 1071 (2007).
178. K.-H. Schweikart, V. L. Malinovskii, J. R. Diers, et al., *J. Mater. Chem.* **12**, 808 (2002).
179. M. Bai, M. Bao, C. Ma, et al., *J. Mater. Chem.* **13**, 1333 (2003).
180. P. Ma, Y. Chen, N. Sheng, et al., *Eur. J. Inorg. Chem.*, No. 7, 954 (2009).
181. K. P. Birin, Y. G. Gorbunova, and A. Y. Tsivadze, *J. Porphyrins Phthalocyanines*, **13**, 283 (2009).
182. K. P. Birin, Y. G. Gorbunova, and A. Y. Tsivadze, *Macroheterocycles* **3**, 210 (2010).
183. K. P. Birin, Y. G. Gorbunova, and A. Y. Tsivadze, *Dalt. Trans.* **40**, 11539 (2011).
184. K. P. Birin, Y. G. Gorbunova, and A. Y. Tsivadze, *Magn. Reson. Chem.* **48**, 505 (2010).